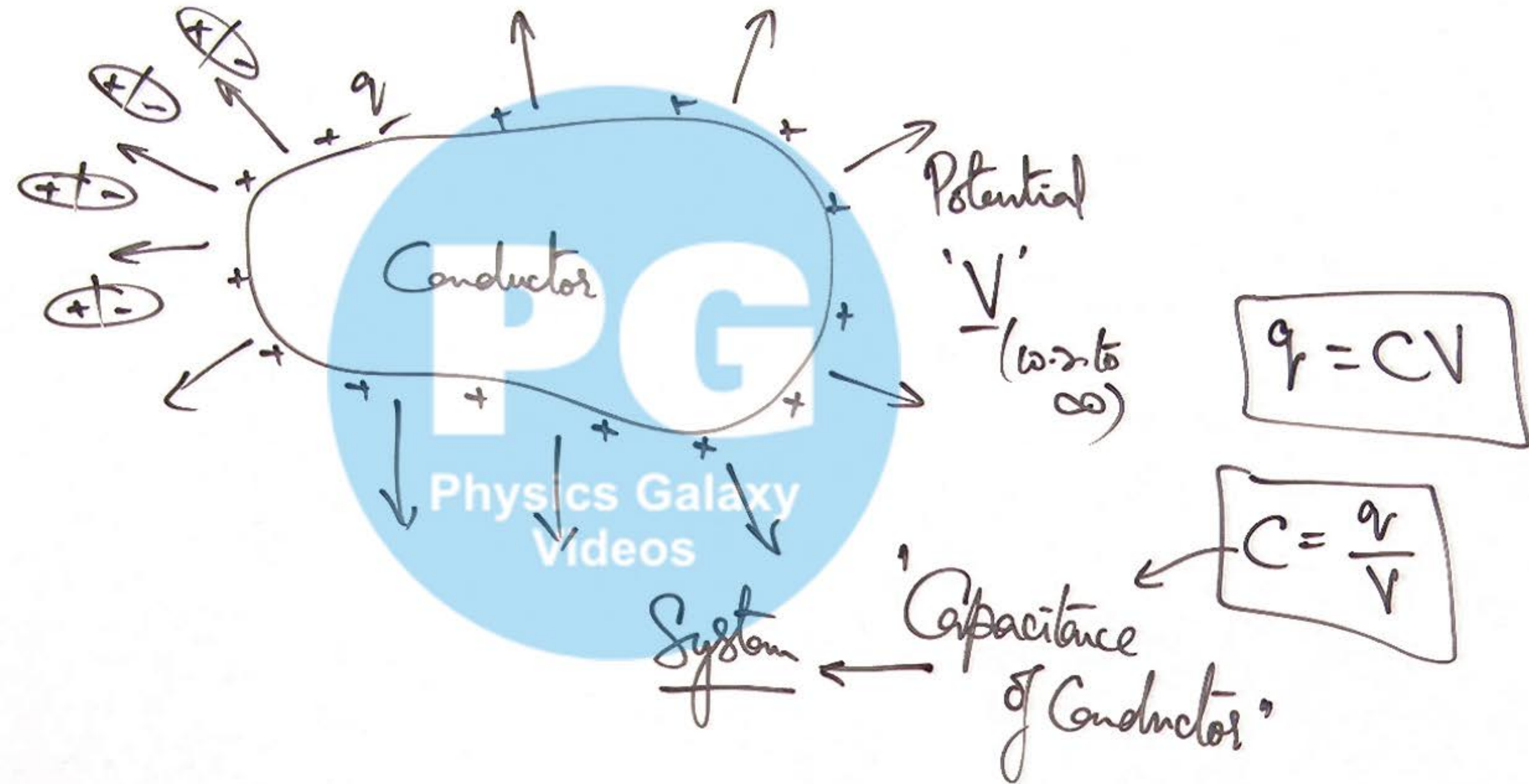
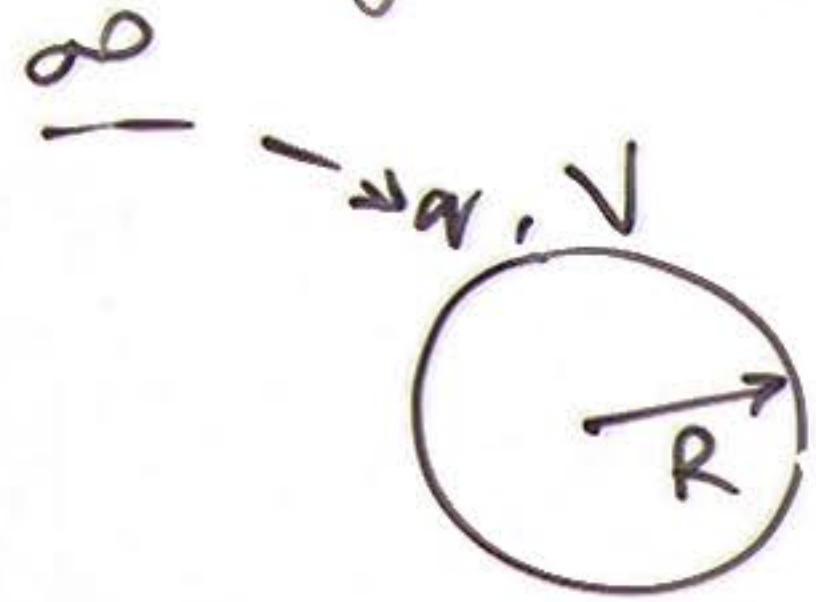


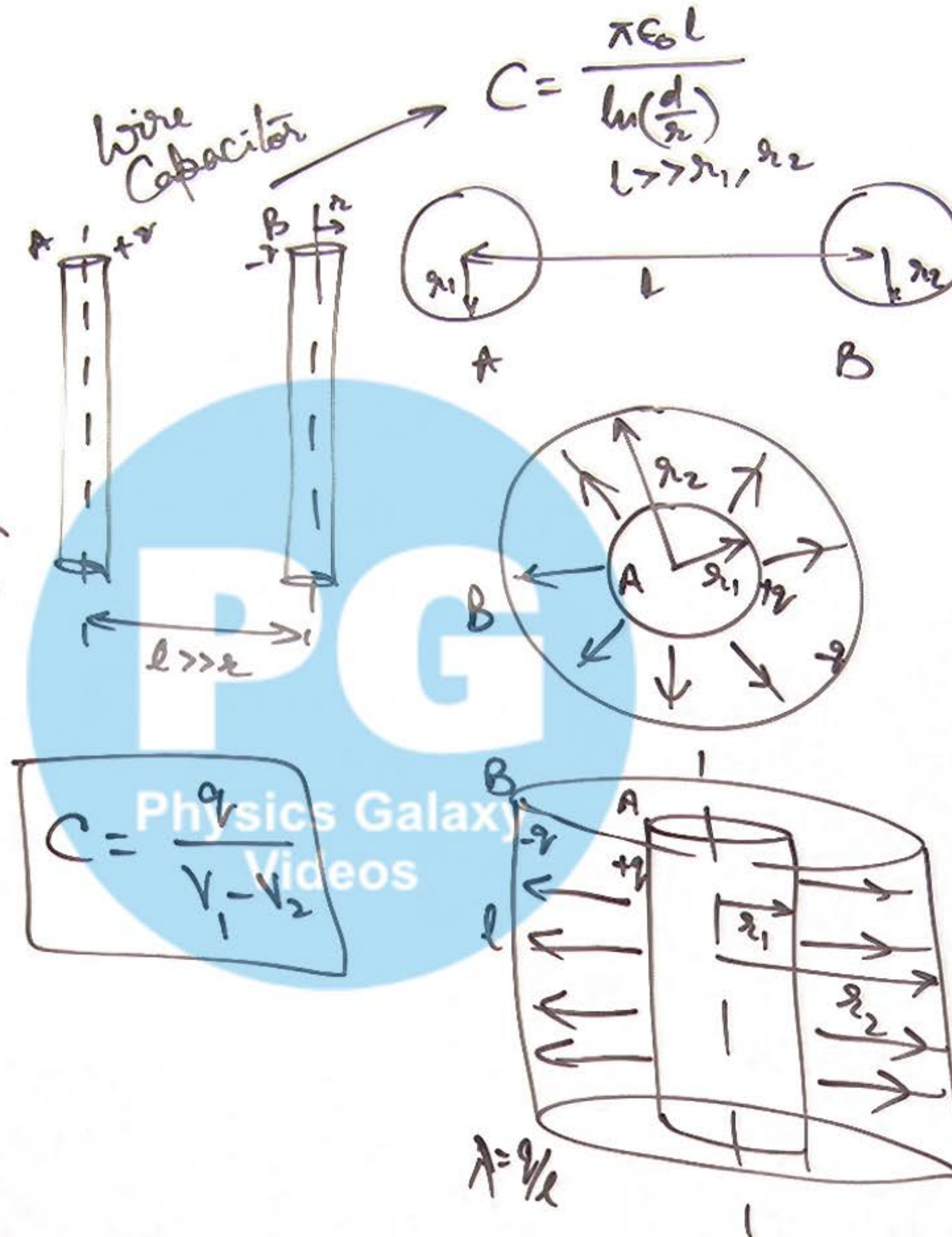
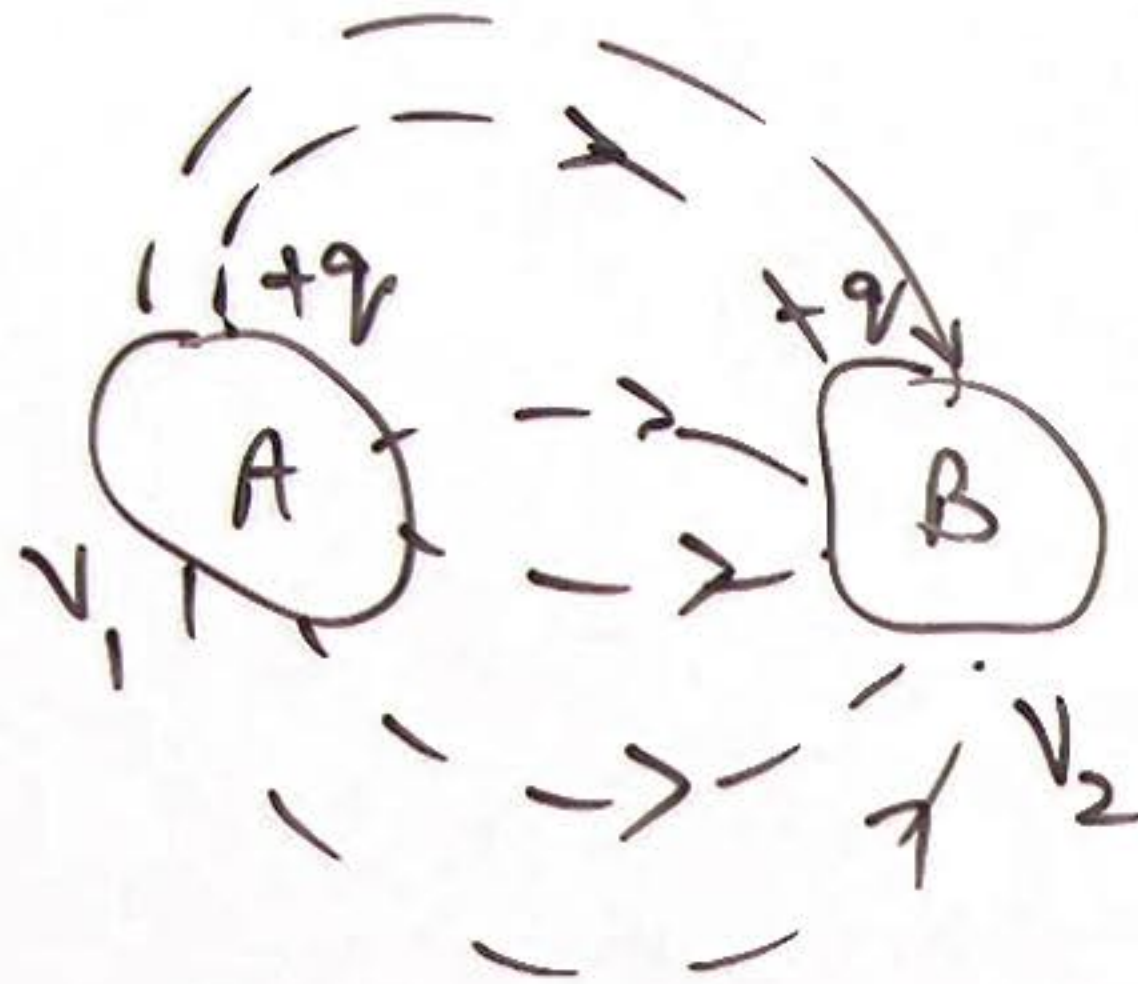
① Capacitance :



② Capacitance of a System of two Conductors:



$$C = \frac{q}{V} = 4\pi\epsilon_0 R$$



Sphere Capacitor

$$C = \left(\frac{4\pi\epsilon_0 r_1 r_2}{r_2 - r_1} \right)$$

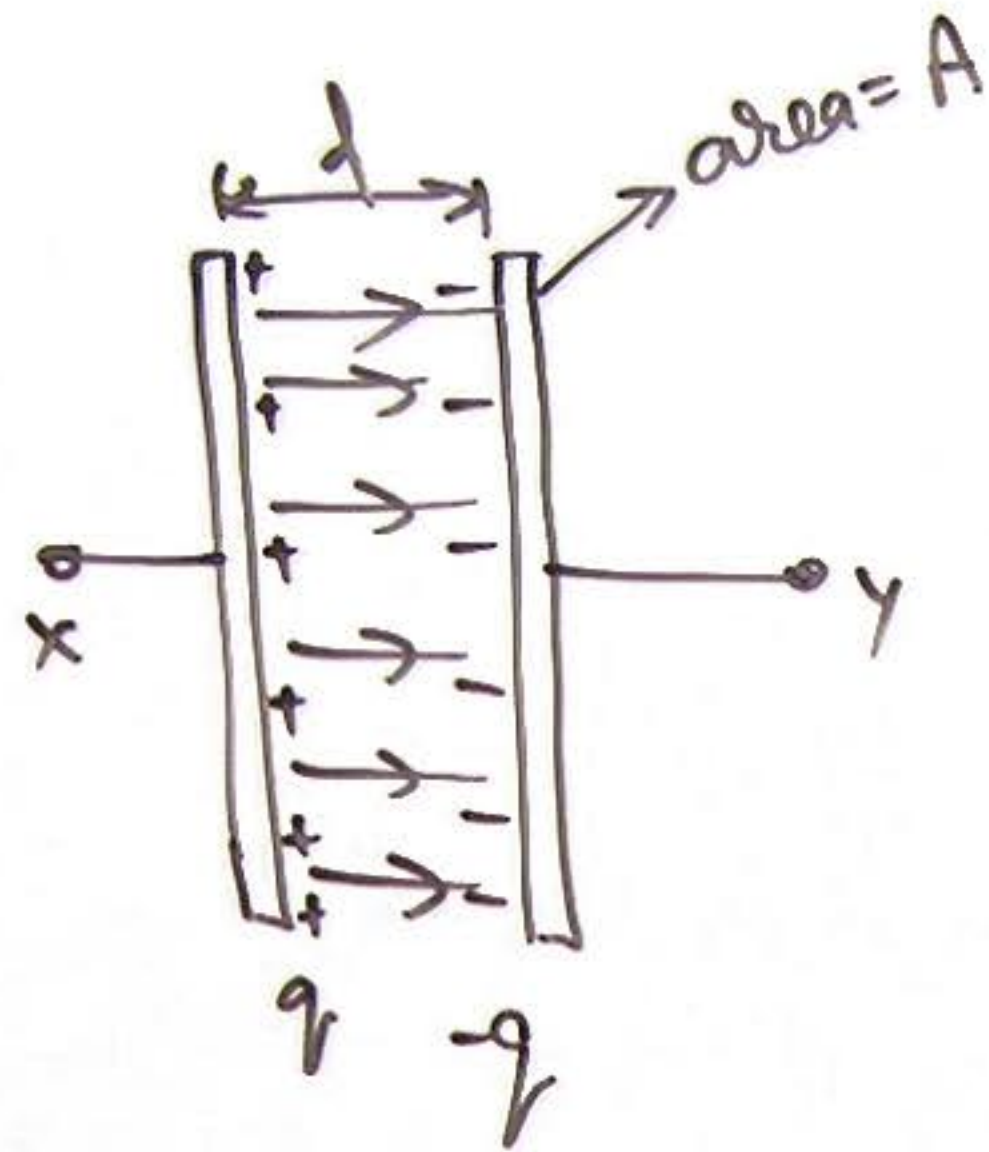
Spherical Capacitor

$$C = \left(\frac{4\pi\epsilon_0 r_1 r_2}{r_2 - r_1} \right)$$

Cylindrical Capacitor

$$C = \frac{2\pi\epsilon_0 l}{\ln(r_2/r_1)}$$

③ Parallel Plate Capacitor:



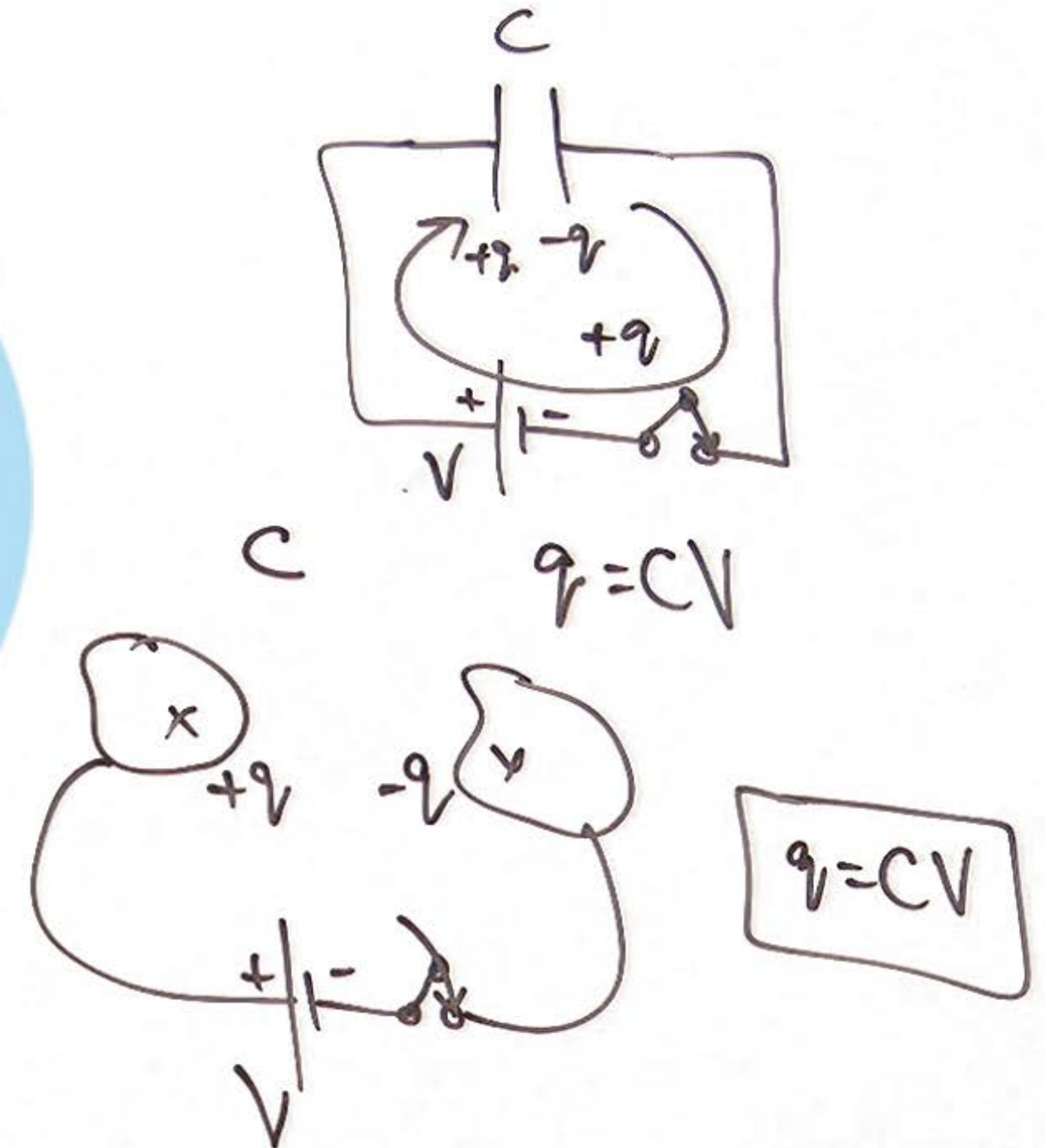
Size of plates $\gg d$

Capacitance $C = \frac{q}{V} = \frac{\epsilon_0 A}{d}$

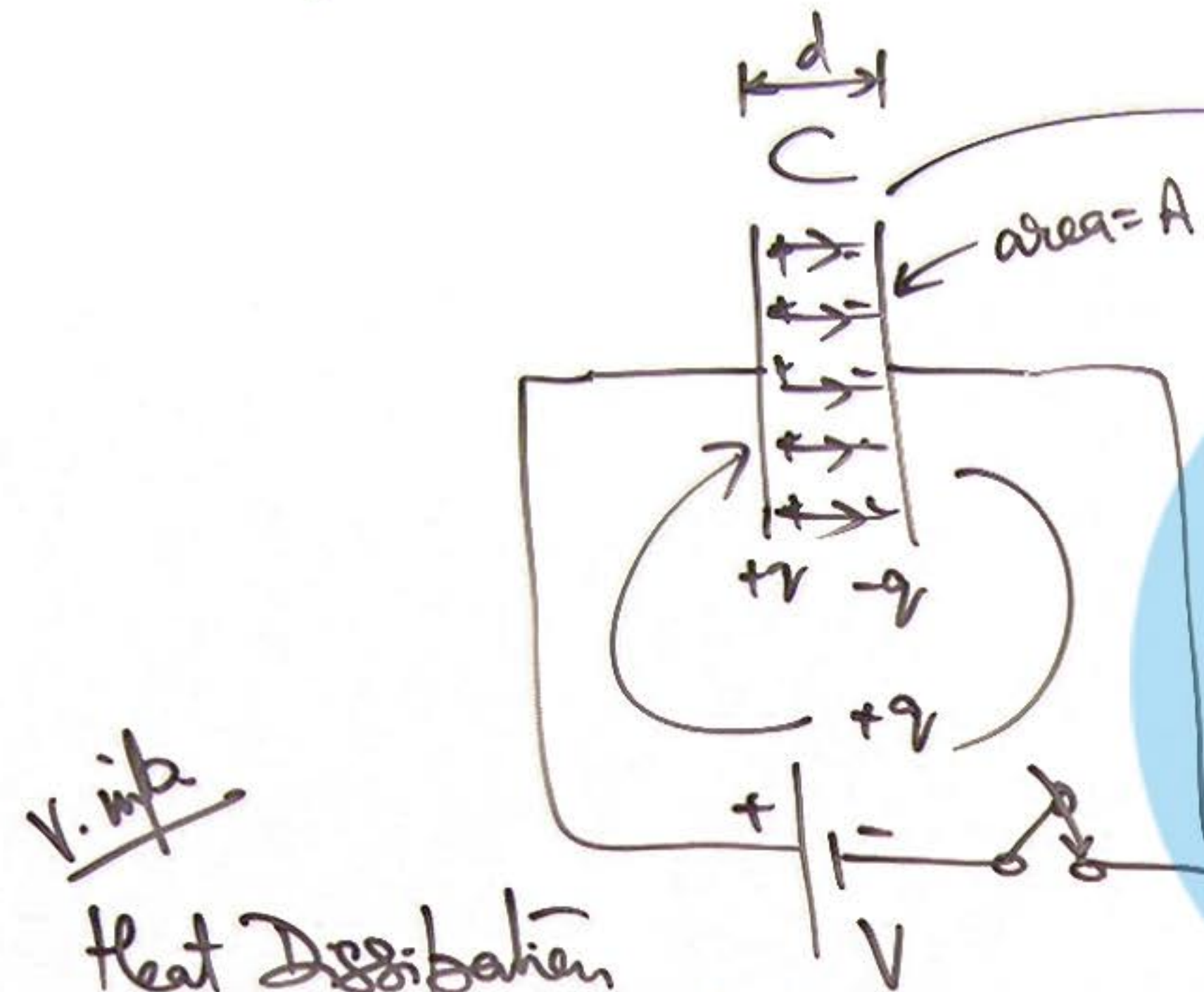
$V = Ed$

$V = \left(\frac{\sigma}{\epsilon_0}\right) d$

$V = \frac{q}{A\epsilon_0} \cdot d$



④ Energy Stored in a Capacitor:



Energy between plates of Cap

$$U = \frac{1}{2} \epsilon_0 E^2 \times Ad = \frac{1}{2} \epsilon_0 \left(\frac{V}{d} \right)^2 \times Ad$$

J/m^3

$$U = \frac{1}{2} CV^2$$

$$q = CV$$

W/D by battery
Physics Galaxy
Videos

$$W = \Delta q \times V$$

$$W = CV^2$$

V. imp.
Heat Dissipation
in charging a group of
N-Capacitors

$$H = \sum_{i=1}^N \frac{\Delta q_i^2}{2C_i}$$

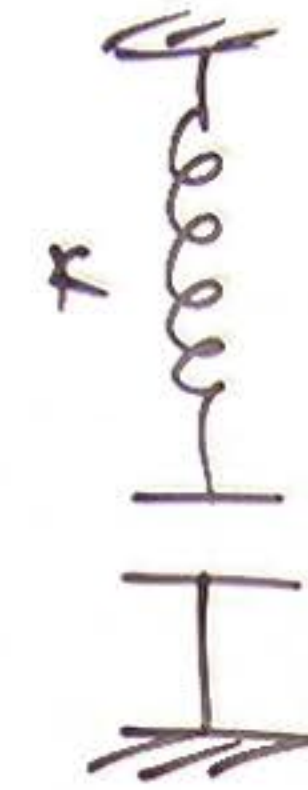
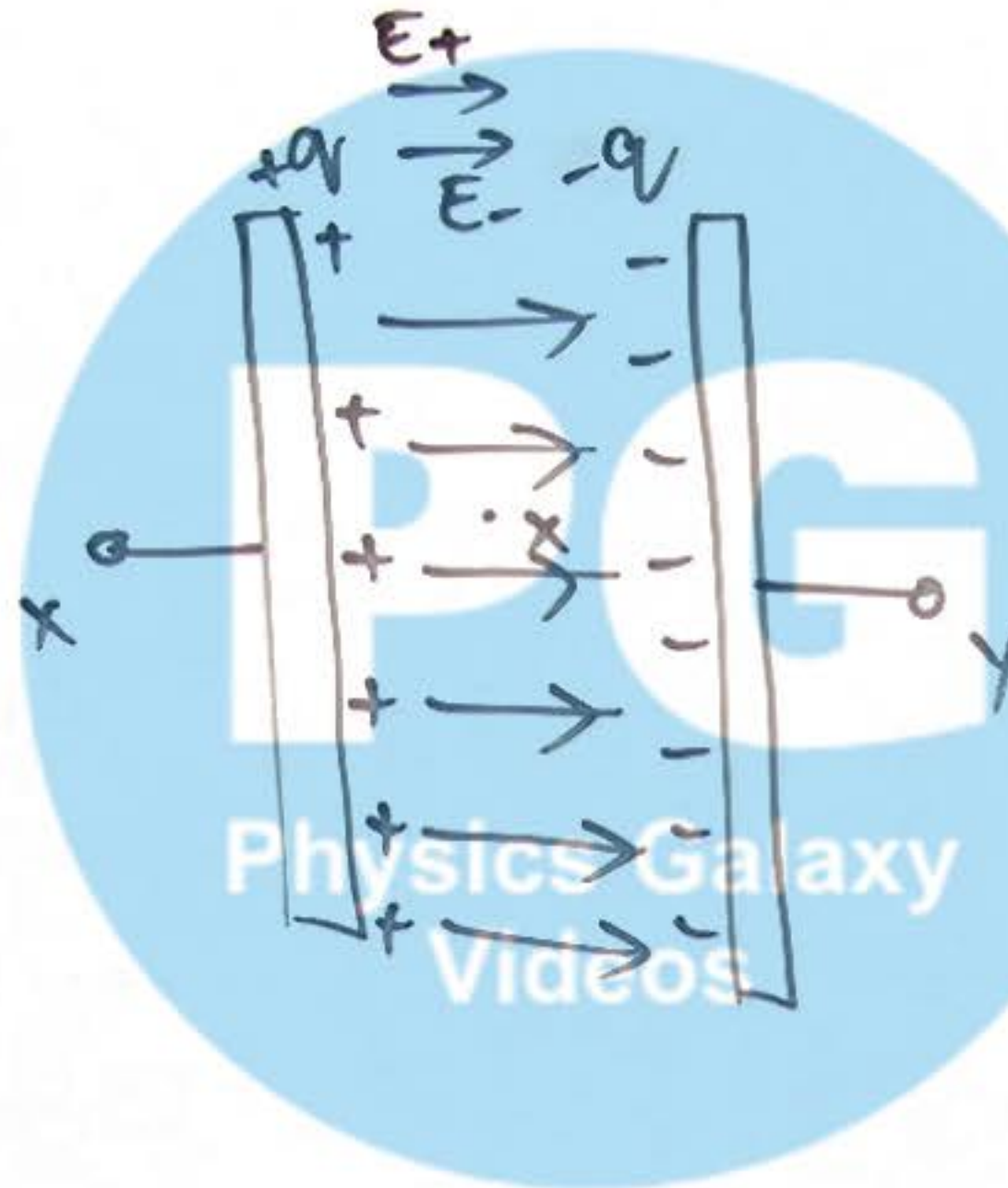
$$\text{Heat dissipation} = W - U$$

$$= CV^2 - \frac{1}{2} CV^2 = \frac{1}{2} CV^2$$

⑤ Force between plates of a Capacitor:

force on one plate due to another

$$F = q|E_+| = \frac{q^2}{2\epsilon_0 A}$$



elongation?

$$\frac{q^2}{2\epsilon_0 A} = kx$$

$x = \checkmark$

$$E_x = \frac{\sigma}{\epsilon_0} = \frac{q}{A\epsilon_0}$$

$$|E_+| = |E_-| = \frac{E_x}{2} = \frac{q}{2A\epsilon_0}$$

E due to any one plate of Capacitor

⑥ Combination of Capacitors :

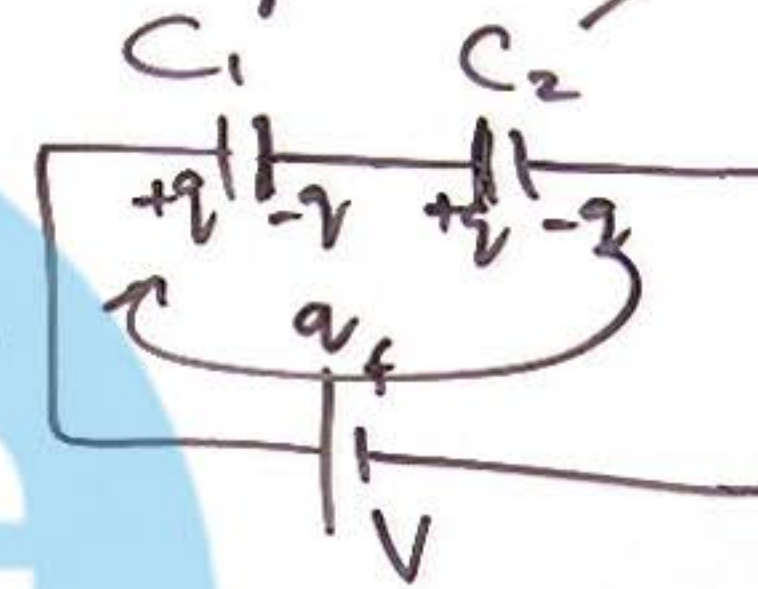
for n-Cap in Combⁿ

$$C_{eq} = \frac{C}{n}$$

$$V_{each} = \frac{V}{n}$$

$q = \frac{C_1 C_2 \cdot V}{C_1 + C_2}$
Same on all cap

Series Combination



$$V_1 = \frac{C_2 V}{C_1 + C_2}$$

$$V_2 = \frac{C_1 V}{C_1 + C_2}$$

$$\frac{C_1}{C_2} = \frac{V_2}{V_1}$$

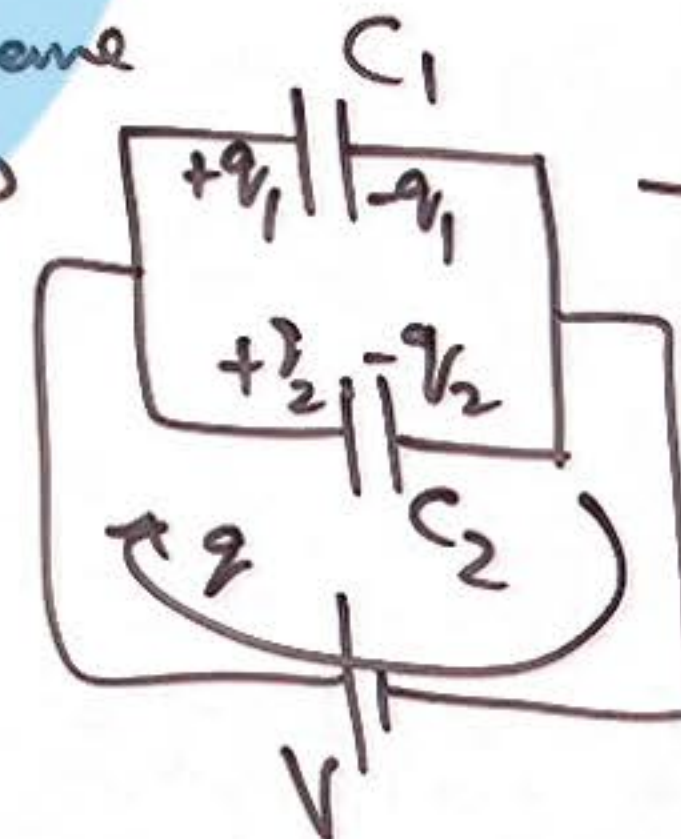
$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$$

$$q_f = q = C_{eq} V$$

Parallel Combination

pd remain same on all cap



$$q_1 = C_1 V$$

$$q_2 = C_2 V$$

$$C_{eq} = C_1 + C_2 + \dots$$

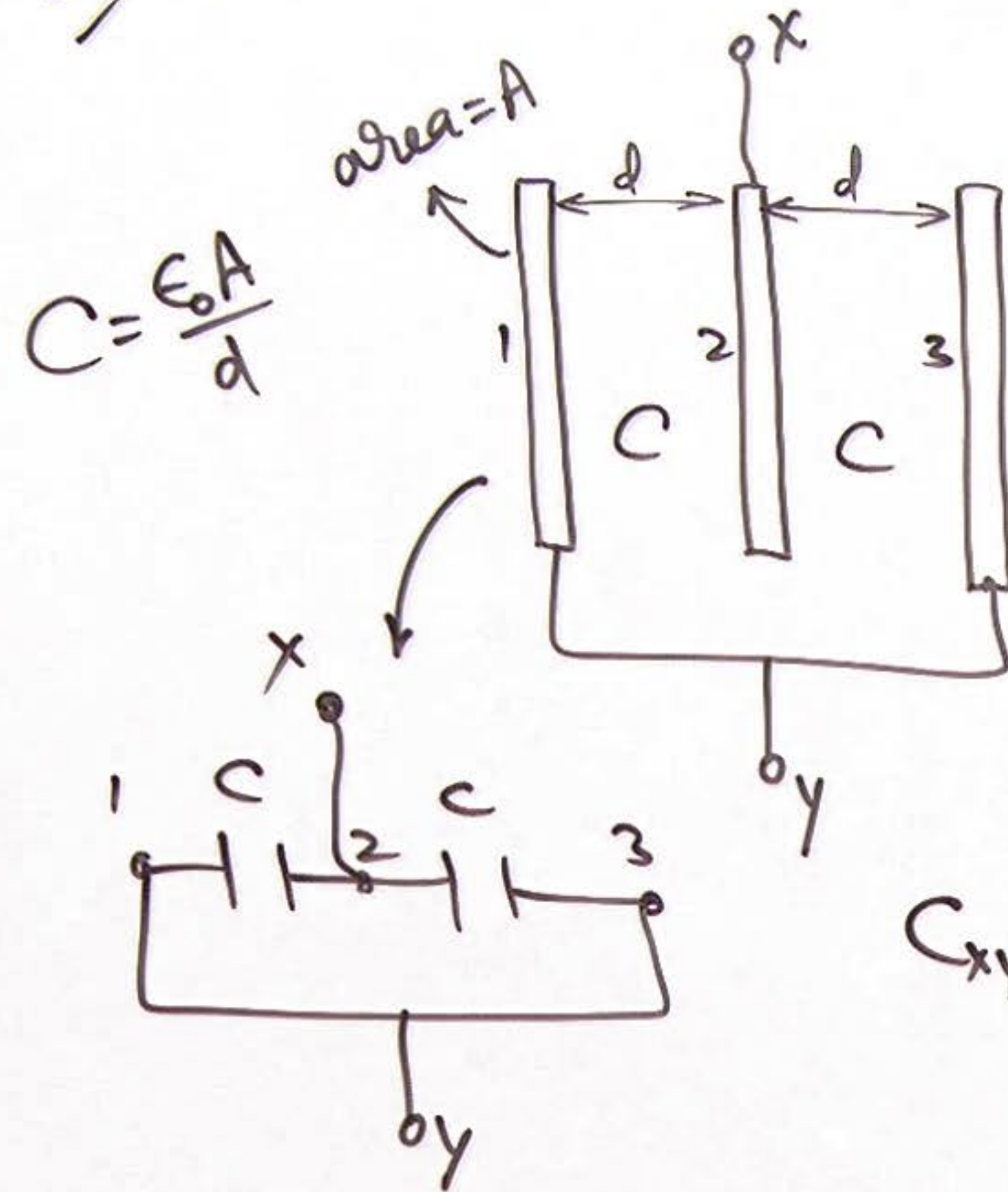
$$q_f = C_{eq} V$$

$$C_{eq} = nC$$

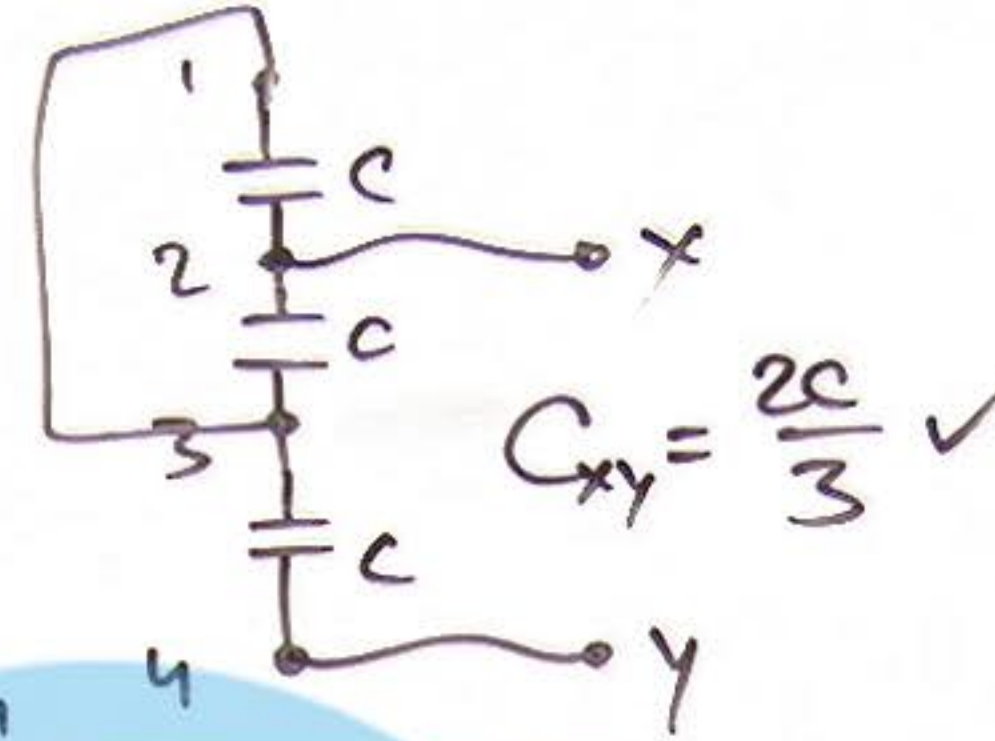
$$V_{each} = V$$

⑦ Combination of parallel Plates:

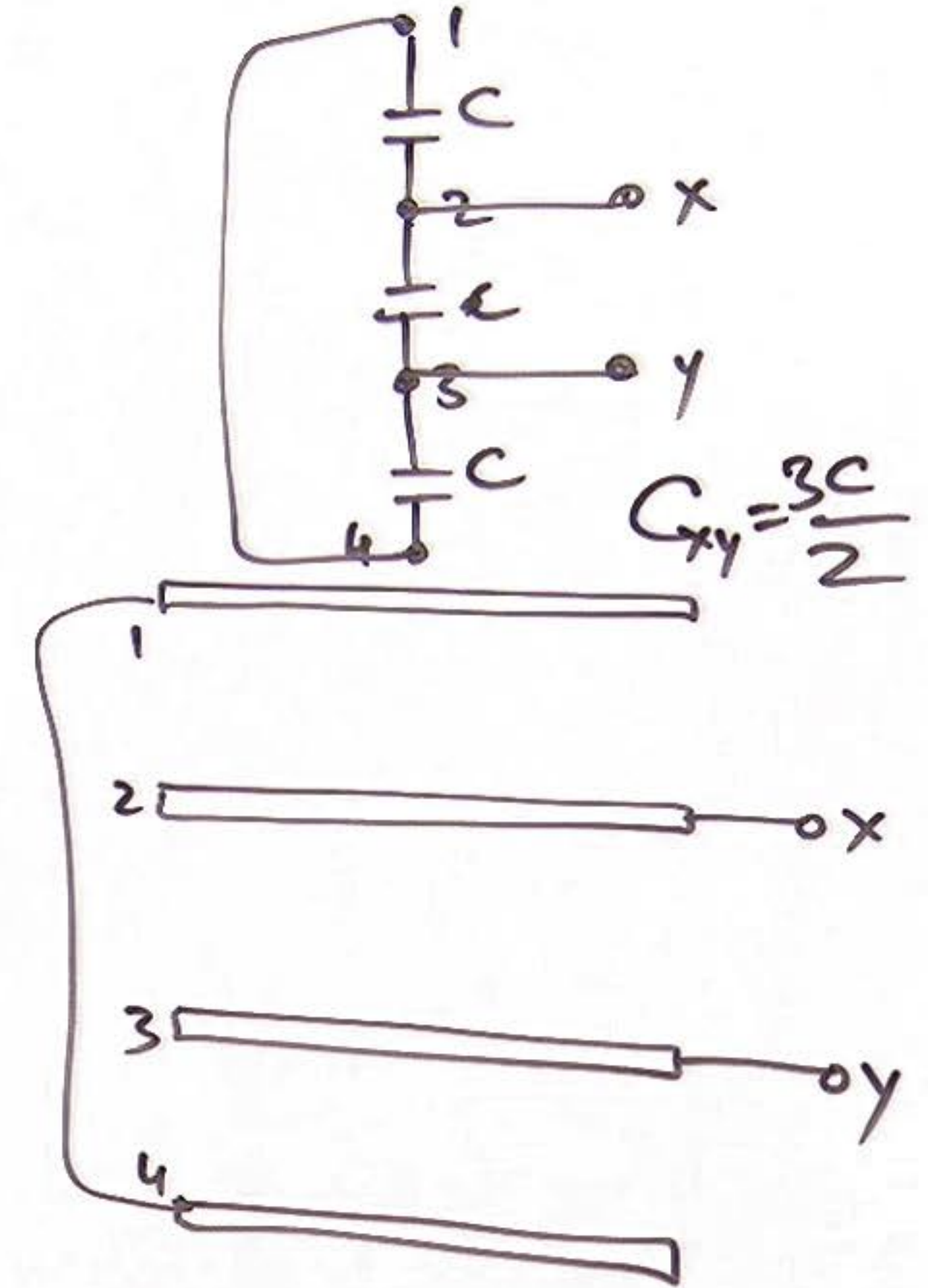
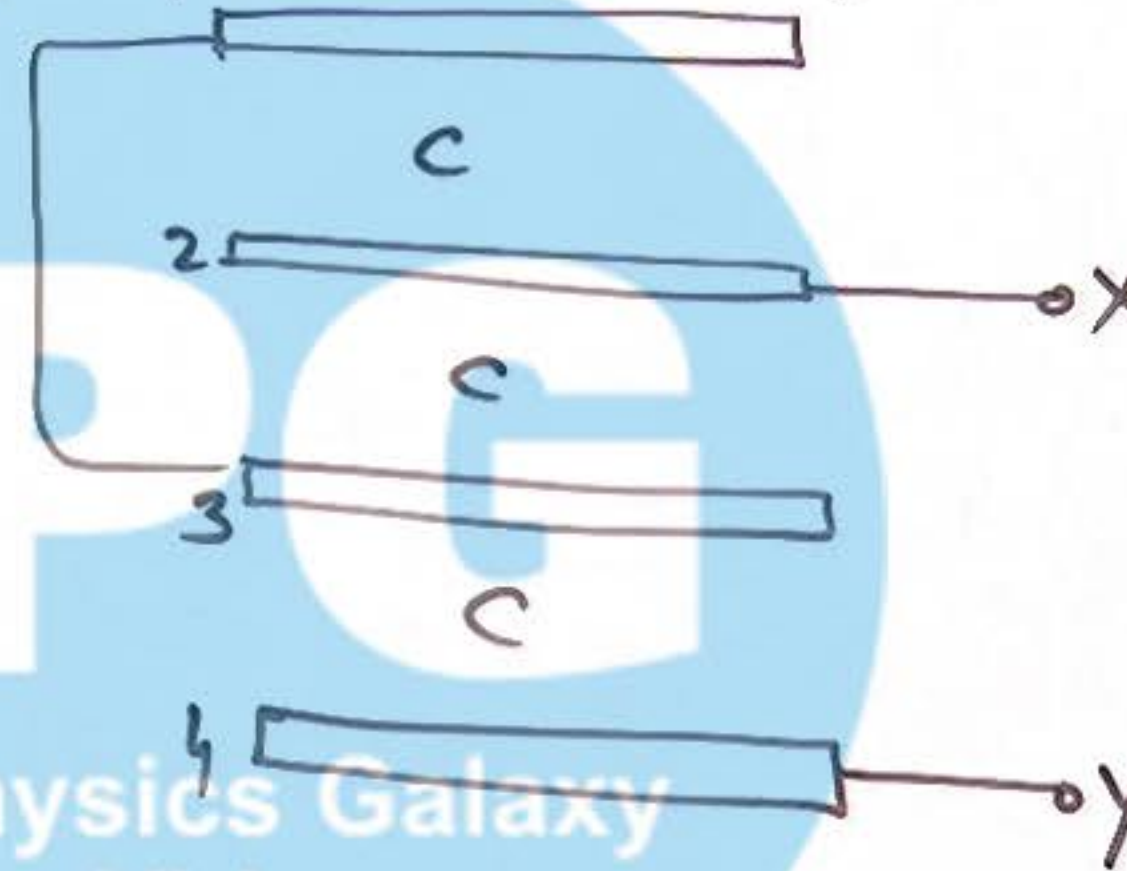
Q1: Find cap across X & Y



$$C_{xy} = 2C$$



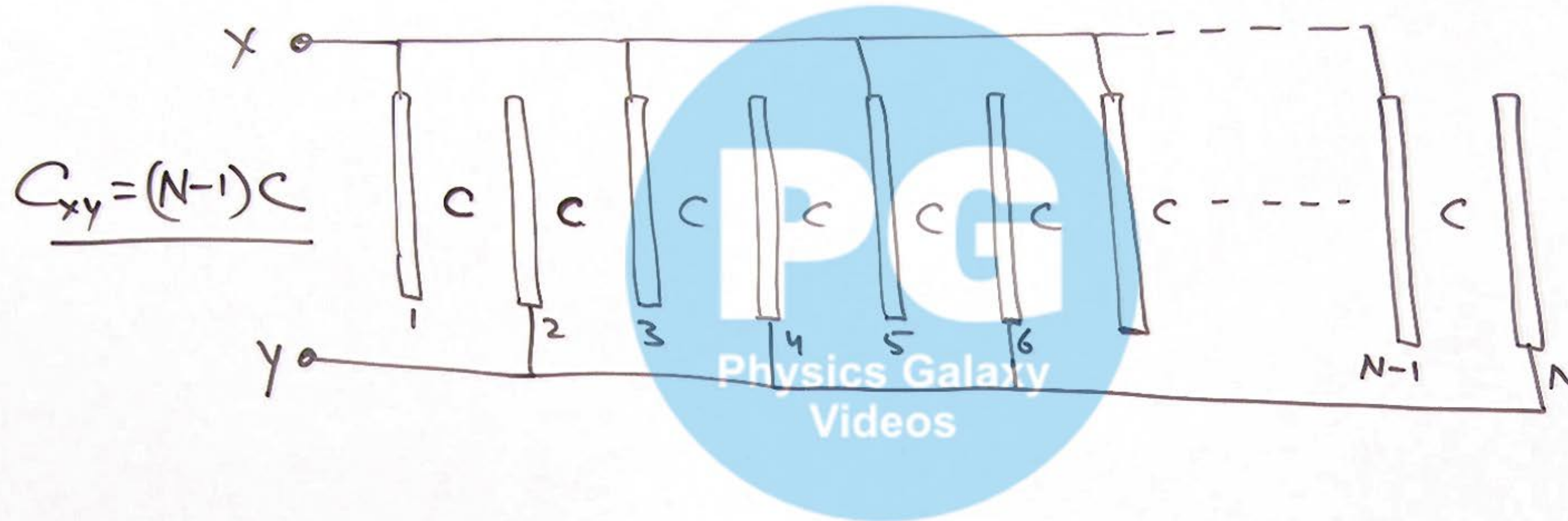
$$C_{xy} = \frac{2C}{3} \checkmark$$



$$C_{xy} = \frac{3C}{2}$$

Physics Galaxy Videos

⑧ Combination of N-Plates:

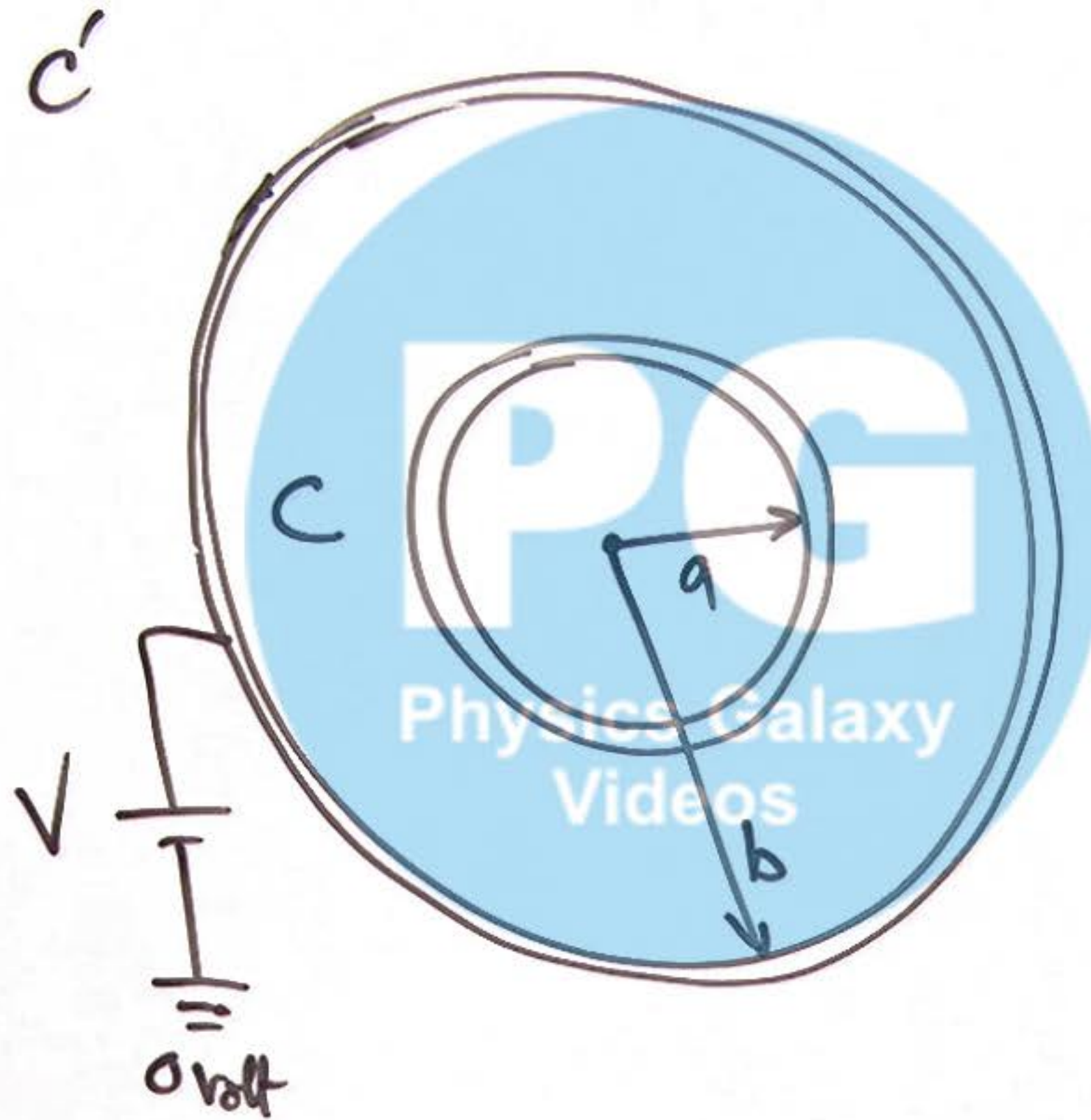


⑨ Types of Spherical Capacitors:

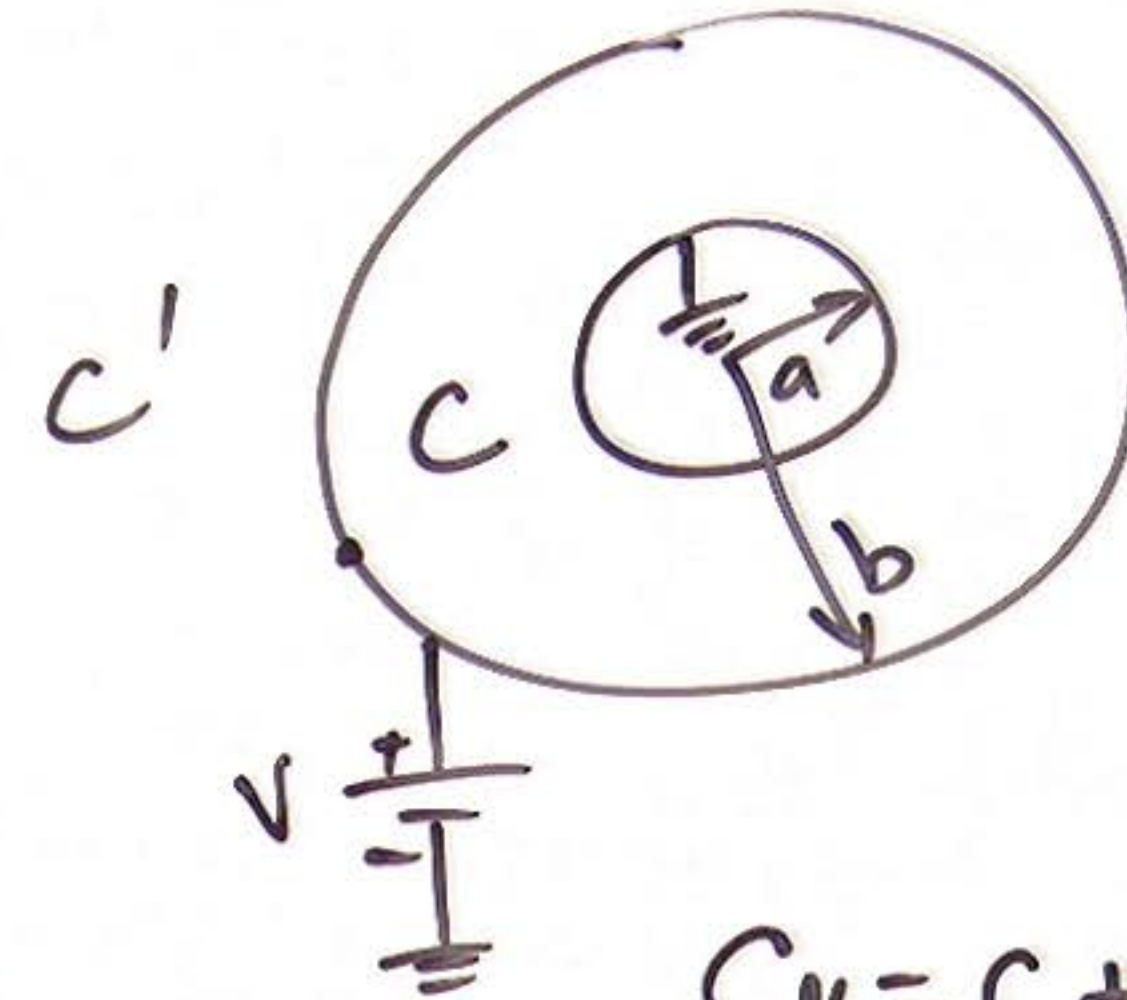
∞ at ∞

$$C = \frac{4\pi\epsilon_0 ab}{b-a}$$

$$C' = 4\pi\epsilon_0 b$$



$V = 0$
 ∞



$$C_{\text{eff}} = C + C'$$

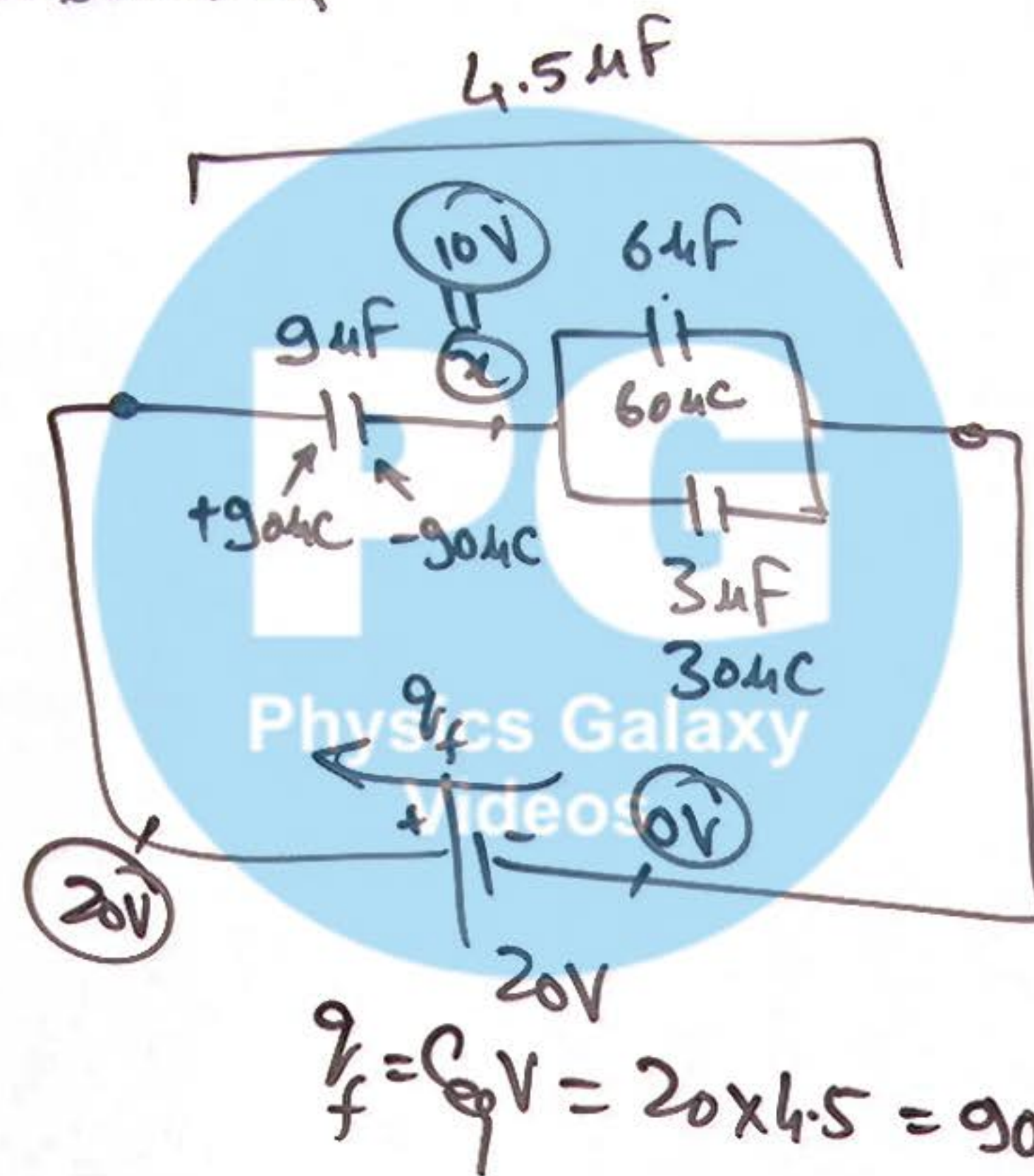
$$= \frac{4\pi\epsilon_0 ab}{b-a} + 4\pi\epsilon_0 b$$

⑩ Capacitive Circuits:

Simpler form $\rightarrow$ S & P combinations

$\rightarrow$ Nodal Analysis

$\downarrow$
method of potential distribution



Nodal eqⁿ for (x)

$$9(x-20) + 6x + 3x = 0$$

$$18x = 180$$

$$x = 10V$$

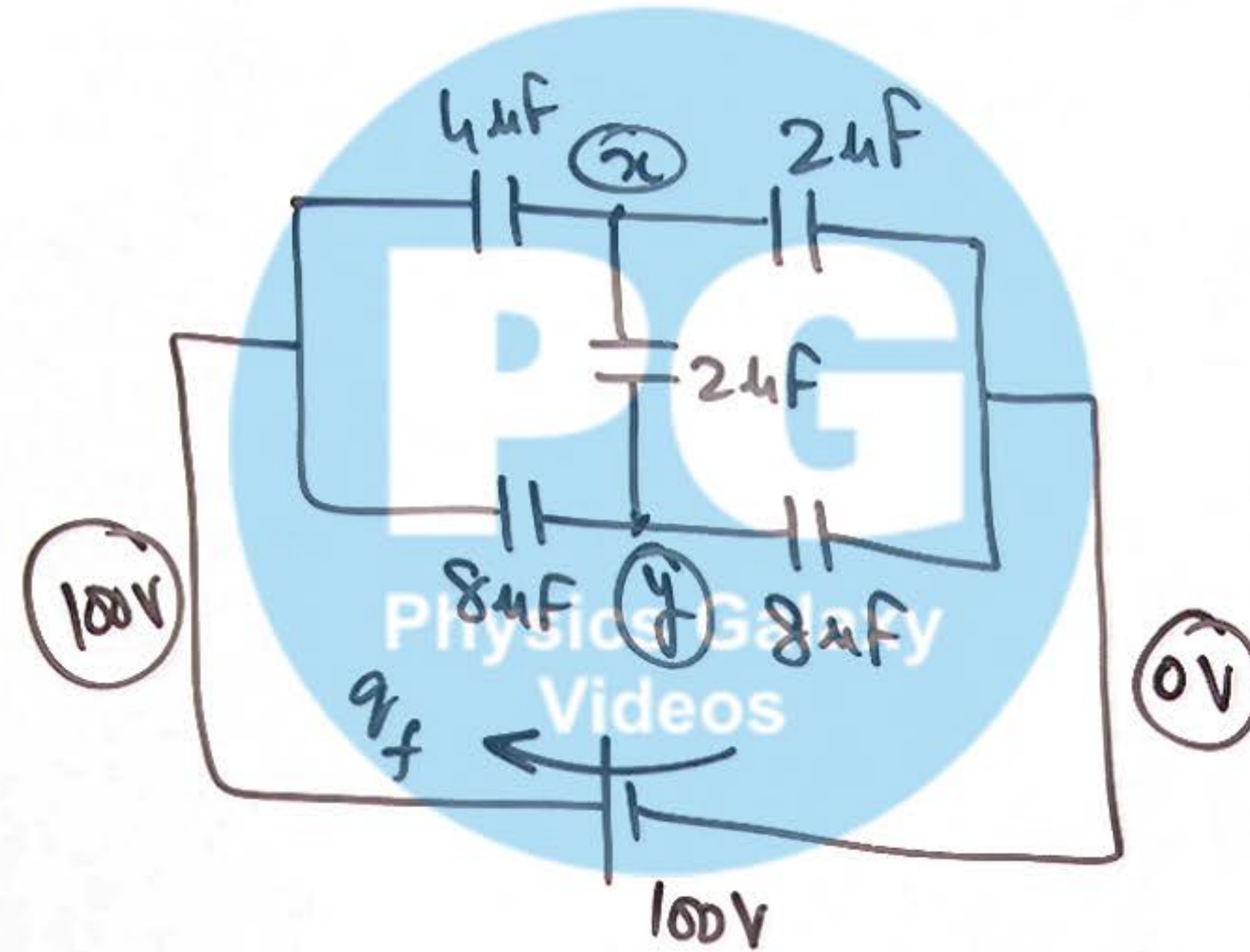
⑩ Capacitive Circuits :

Simpler form

→ Nodal Analysis

↓
method of potential distribution

$$C_{eq} = \frac{q_f}{100} = \frac{2x + 8y}{100} = \underline{\hspace{2cm}}$$



Nodal eqⁿ for (x)

$$2x + 2(x - y) + 4(x - 100) = 0$$

for (y)

$$8y + 2(y - x) + 8(y - 100) = 0$$

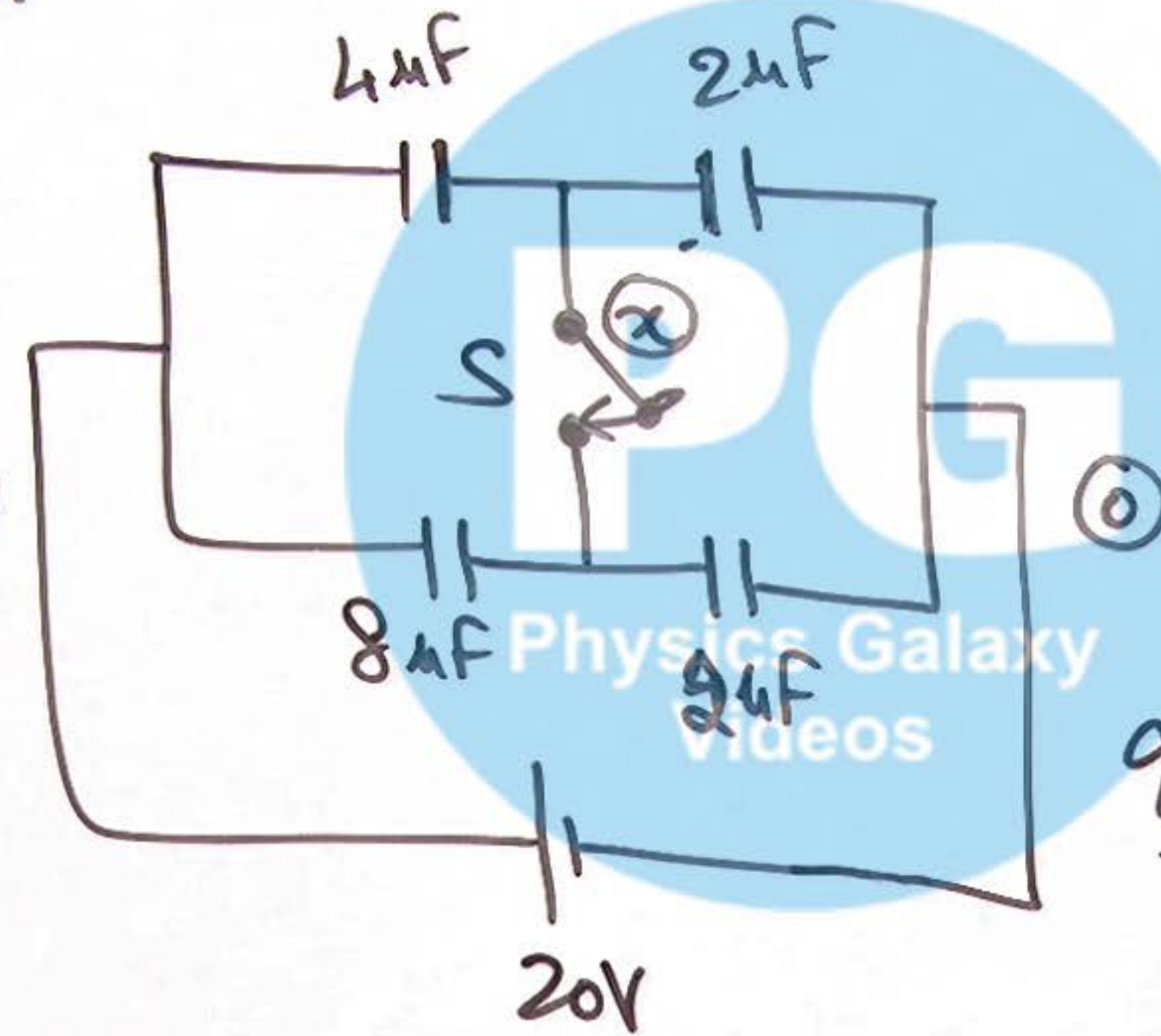
Solving $x = \underline{\hspace{2cm}}$
 $y = \underline{\hspace{2cm}}$

⑪ Switching Circuits:

IL: Find charge flown through switch when it is closed.

$$\text{Heat} = \sum_{i=1}^4 \frac{\Delta q_i^2}{2C_i} \quad \text{②0}$$

= ✓



$$2x + 2x + 4(x-20) + 2(x-20) = 0$$

$$8x = 120$$

$$x = \frac{120}{8} = 15V$$

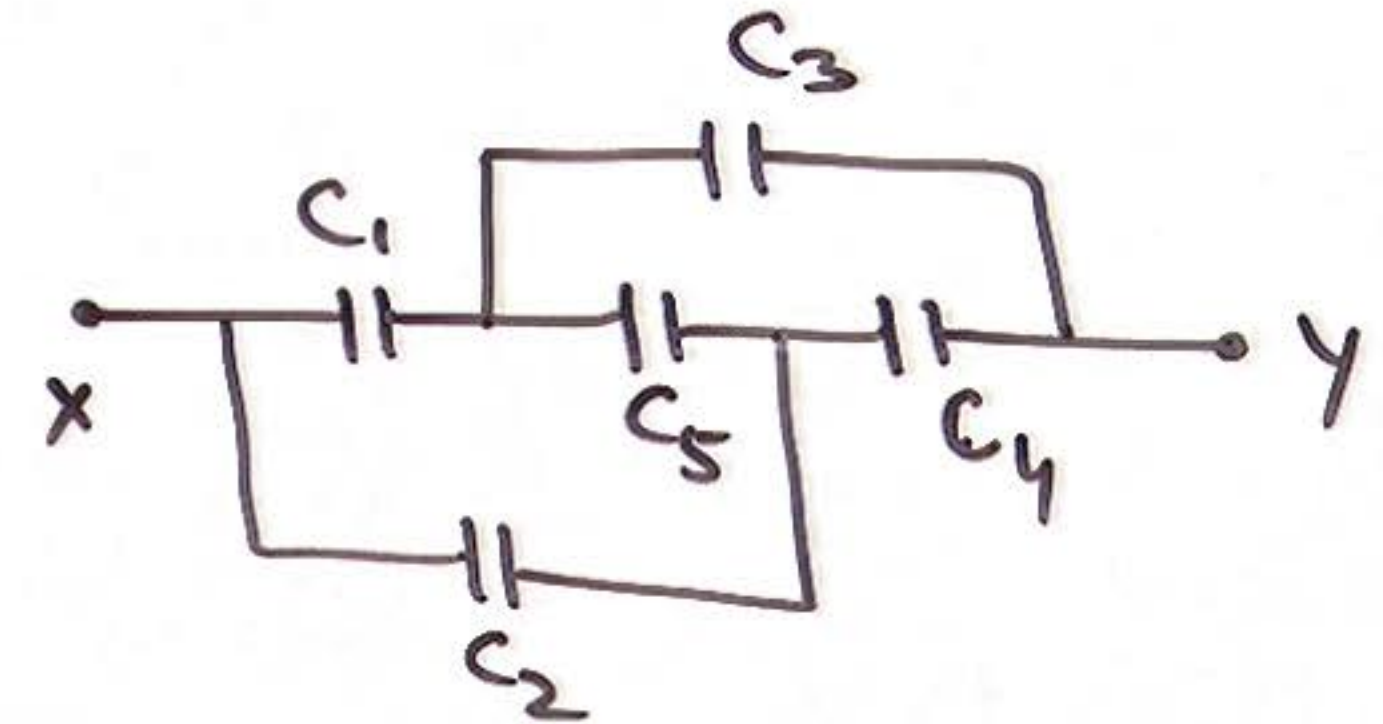
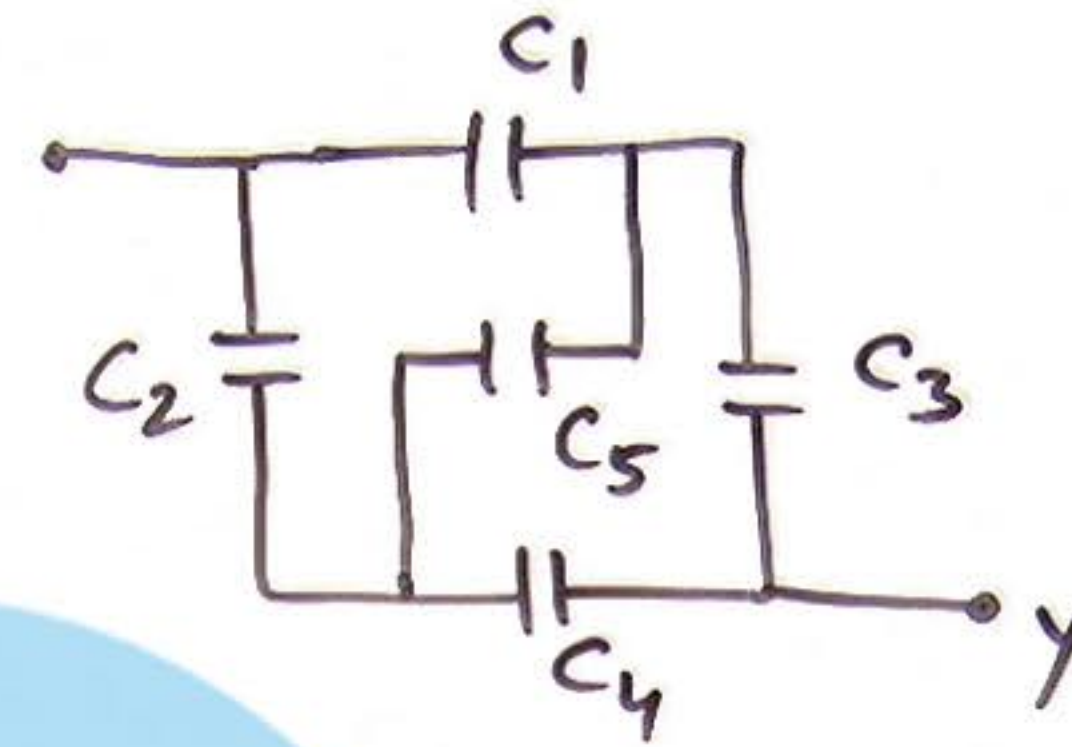
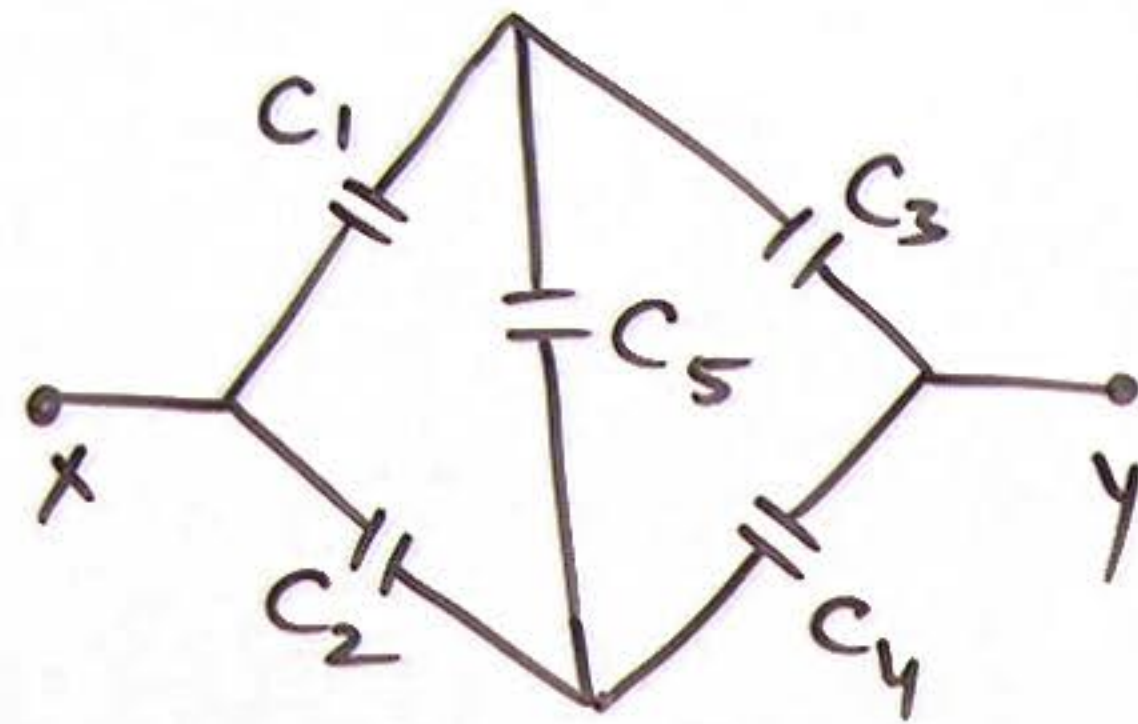
$$q_s = 4(15-20) + 2 \times 15$$

$$= -20 + 30 = +10 \mu C \text{ } q_s$$

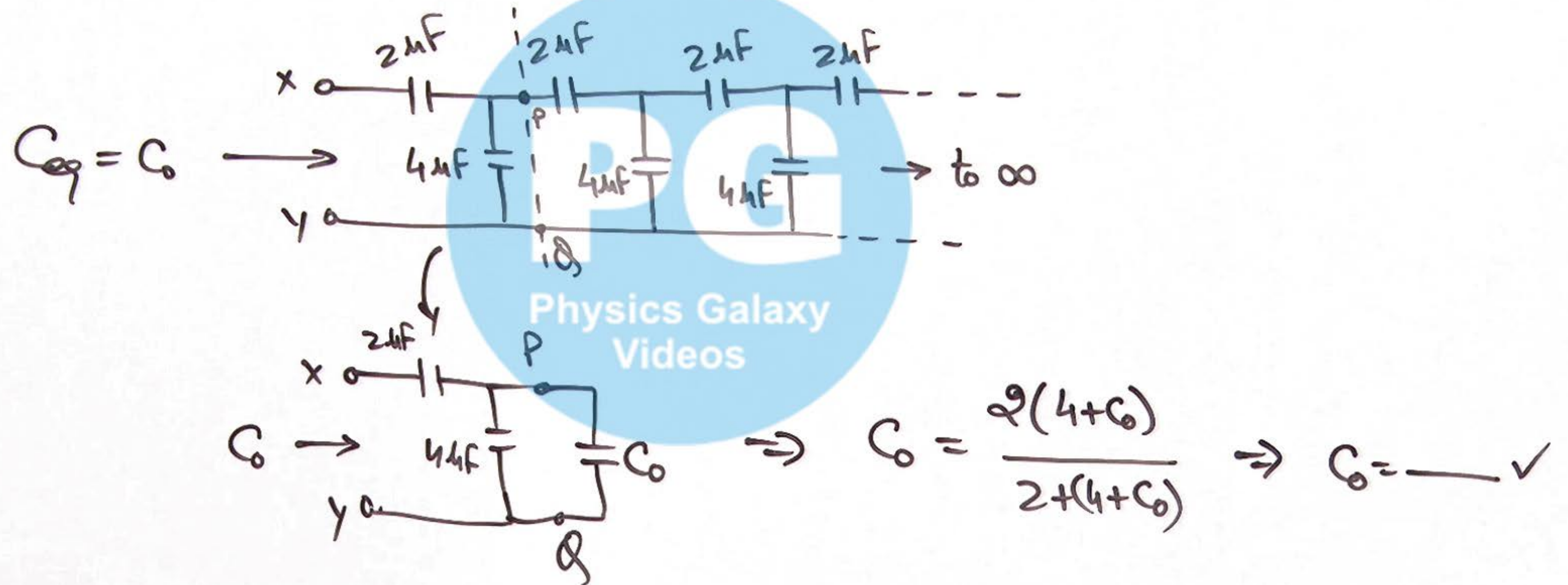
12) Different ways to draw a Wheatstone Bridge:

Balancing Condition

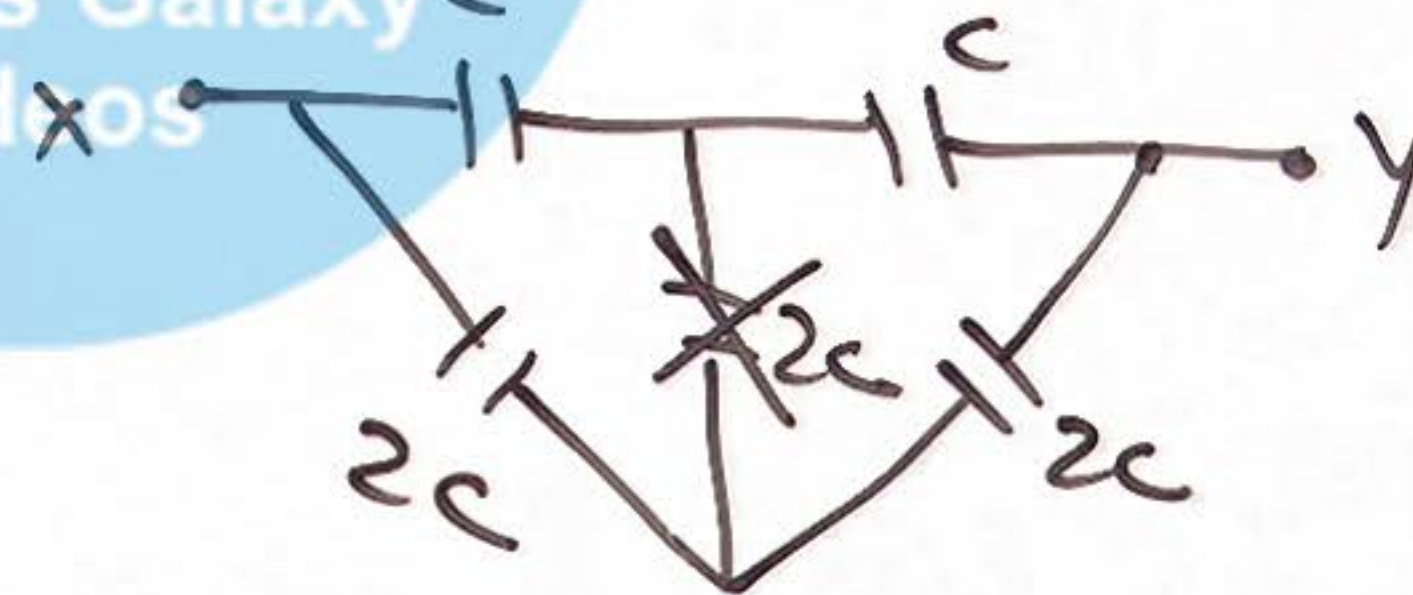
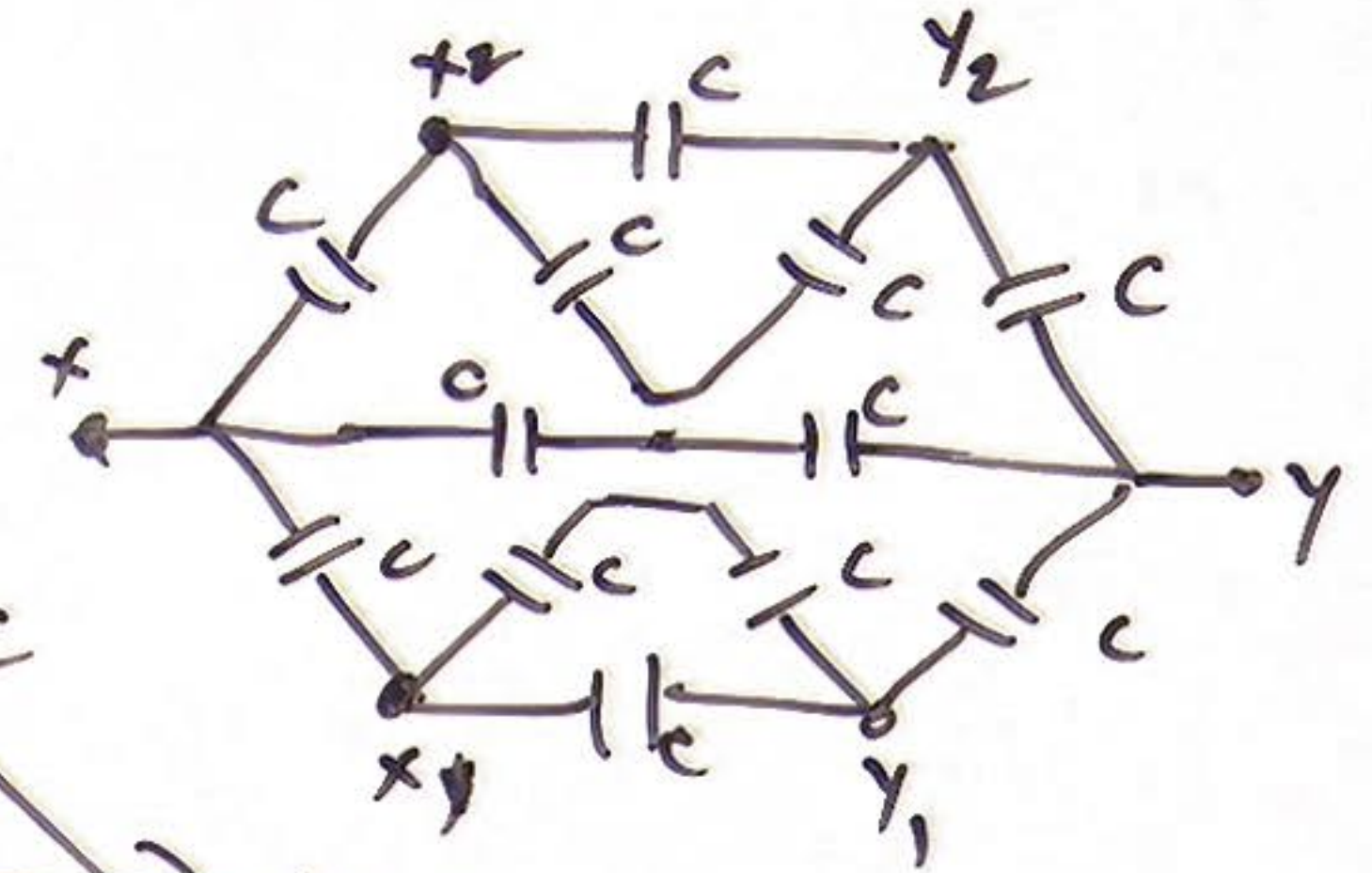
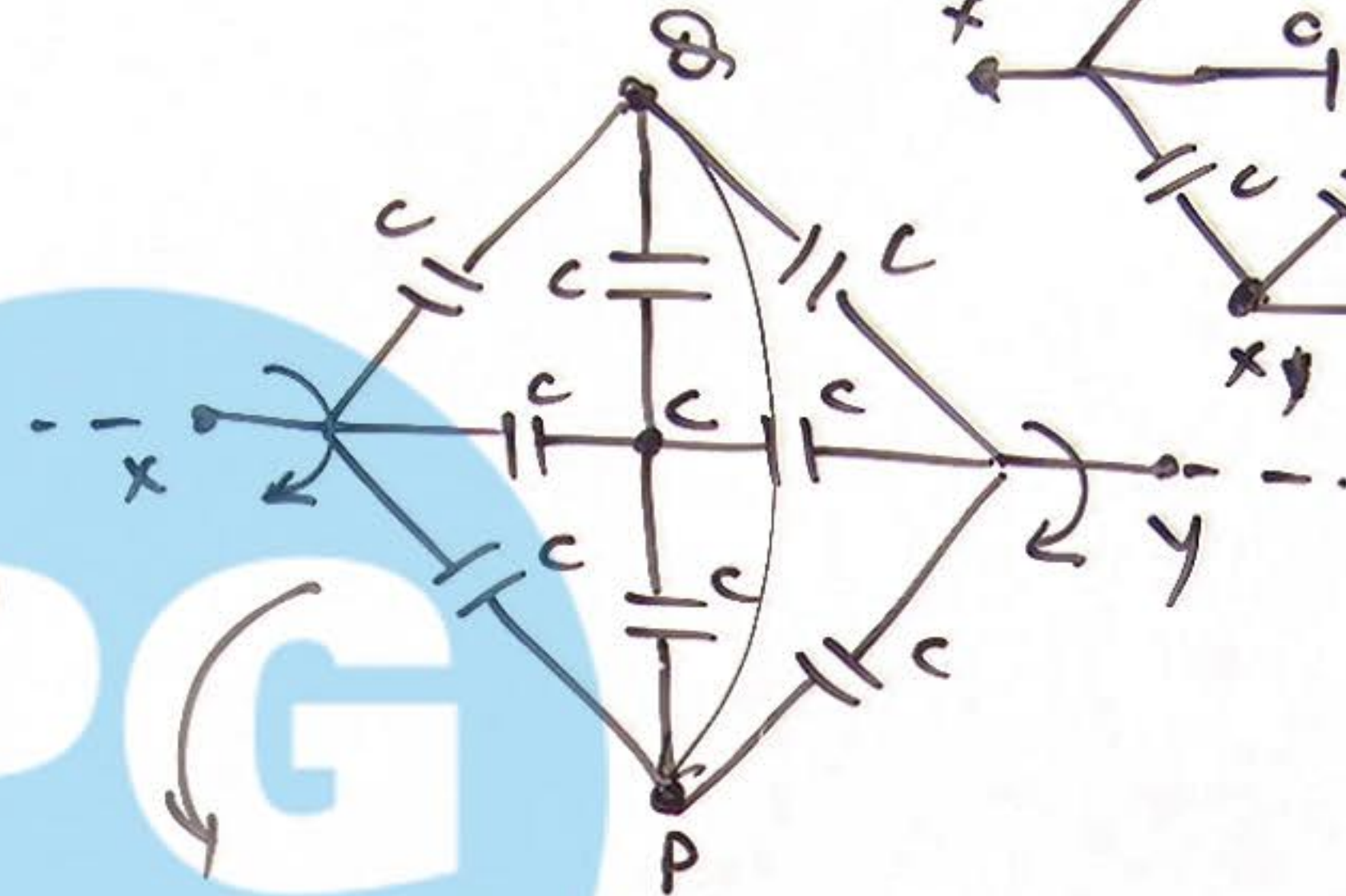
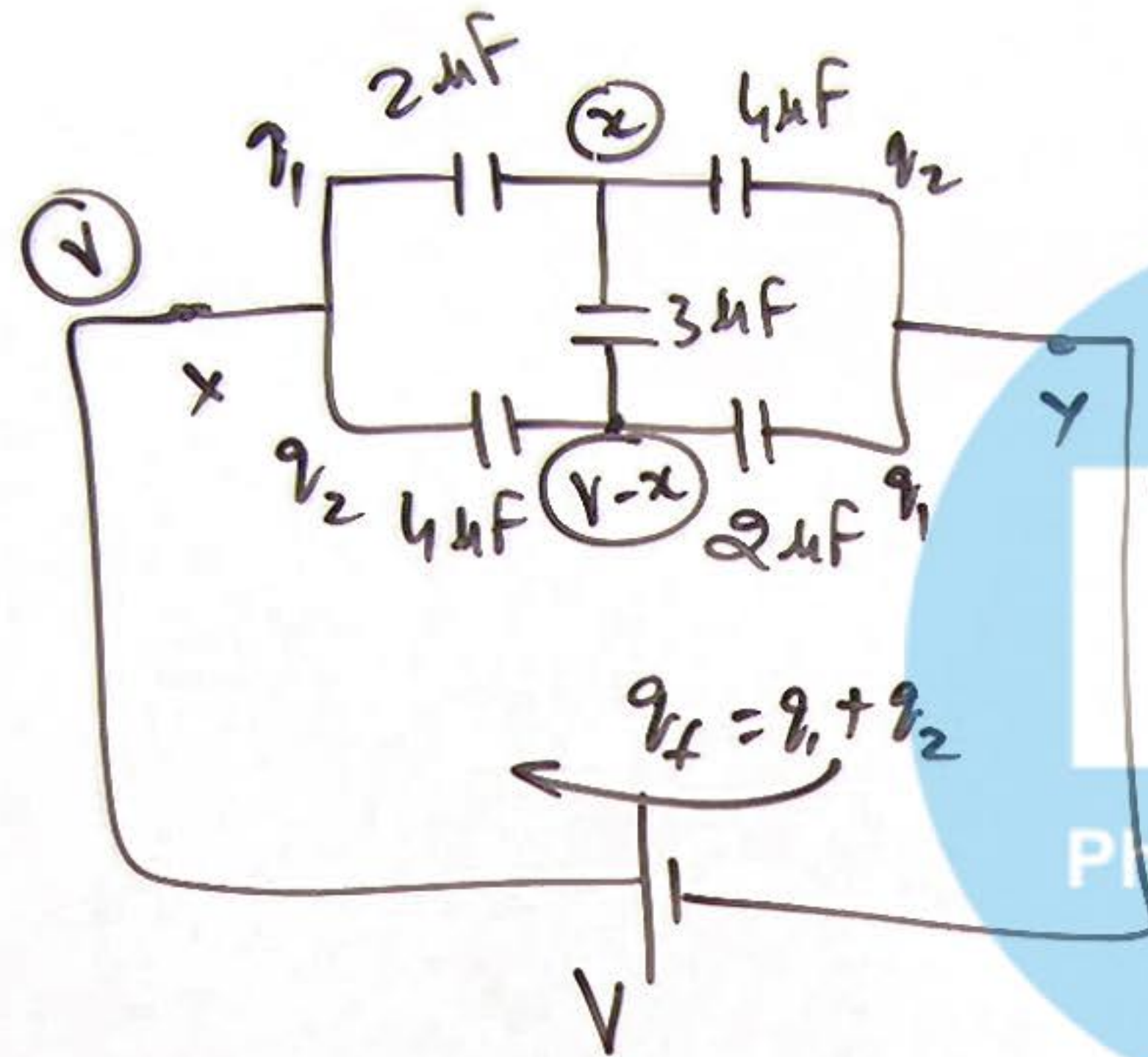
$$\frac{C_1}{C_2} = \frac{C_3}{C_4}$$



⑬ Ladder Networks:

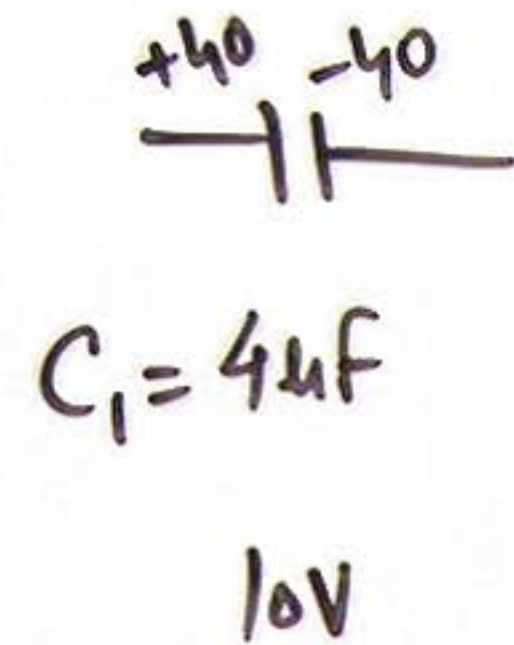


14 Symmetry Circuits:



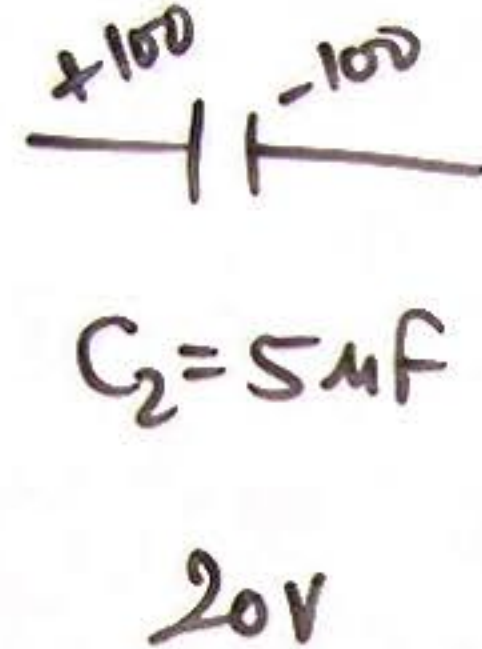
$$C_{xy} = \checkmark$$

⑮ Charge Distribution between Capacitors:



$$C_1 = 4\mu F$$

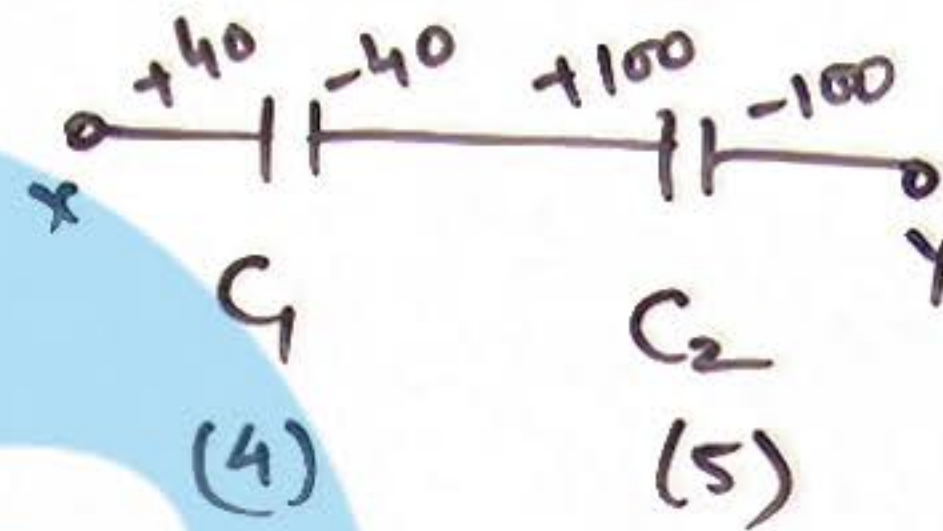
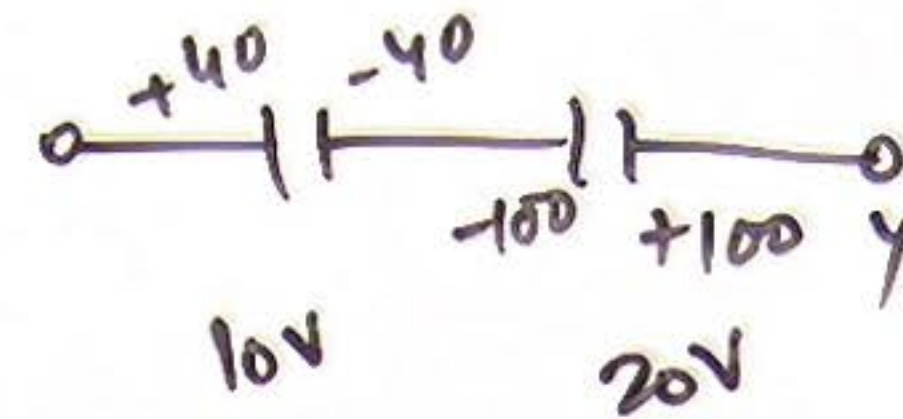
$$10V$$



$$C_2 = 5\mu F$$

$$20V$$

Series



$$V_x - 10 + 20 = V_y$$

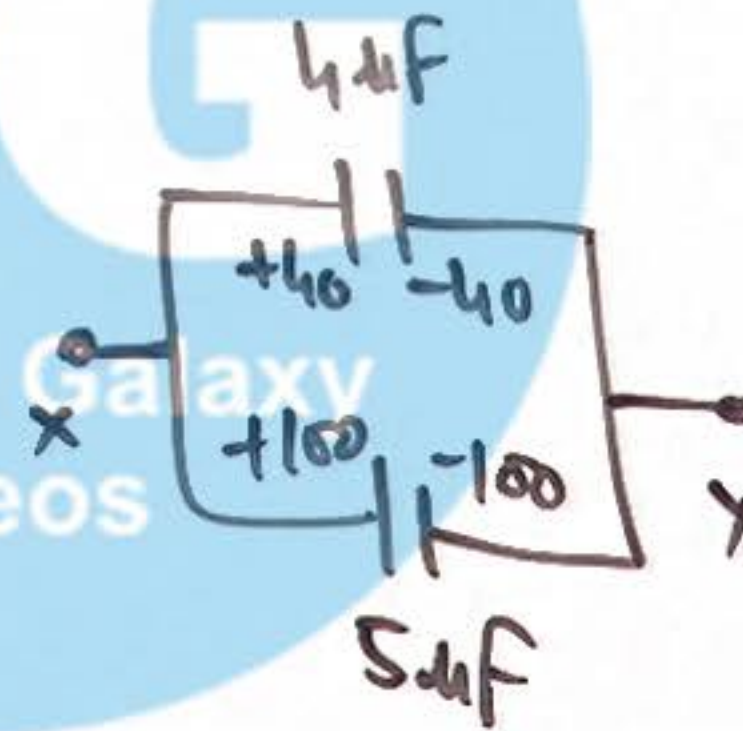
$$V_x - V_y = -10V$$

$$V_x - 10 - 20 = V_y$$

$$V_x - V_y = 30V$$

Parallel

Physics Galaxy
Videos



$$V_x - V_y = \frac{140}{9} V$$

$$q_{\text{new } 4\mu F} = 4 \times V_{xy}$$

$$q_{\text{new } 5\mu F} = 5 \times V_{xy}$$

①⑥ Dielectrics in Capacitors:

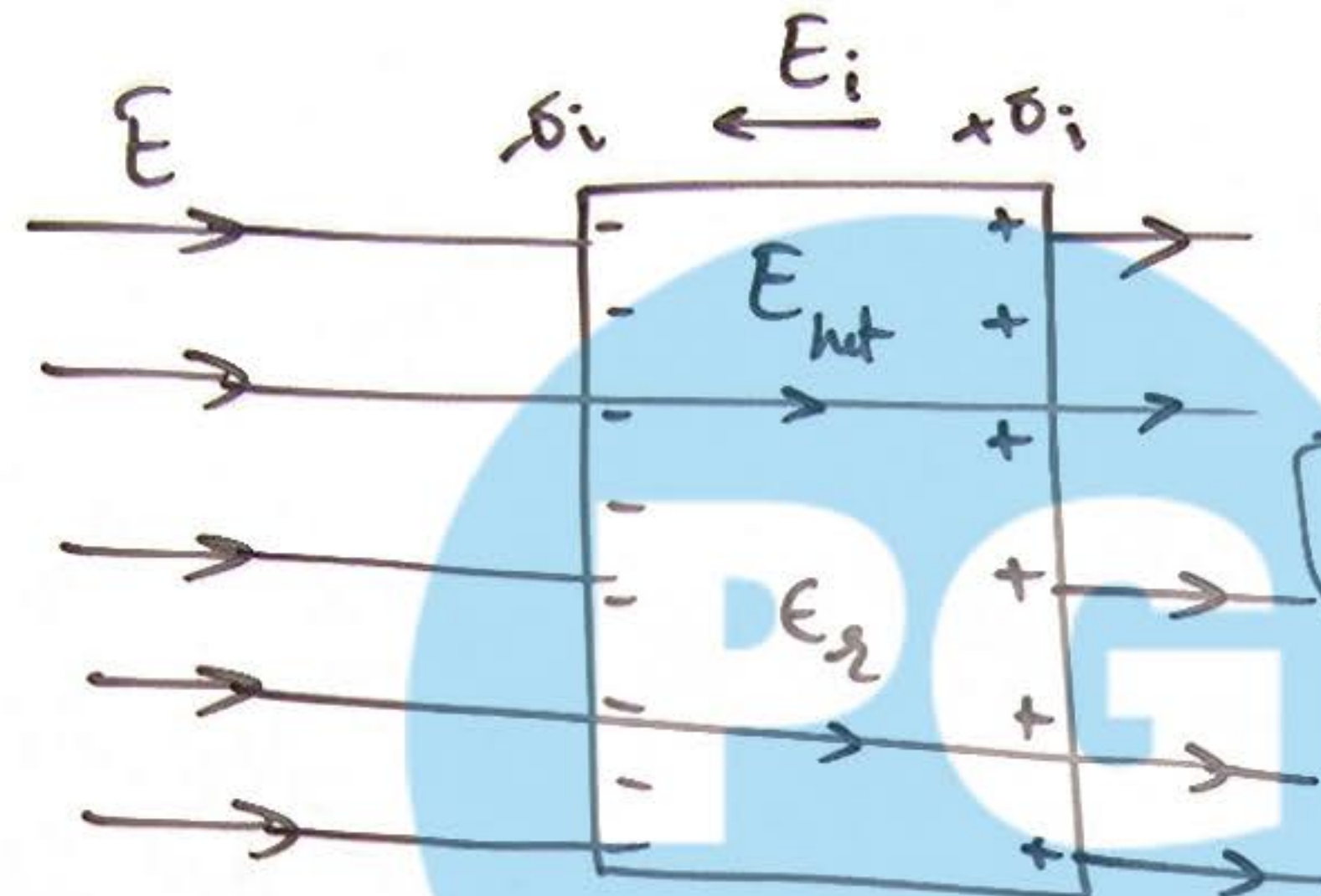
Insulators

Dielectric Constant

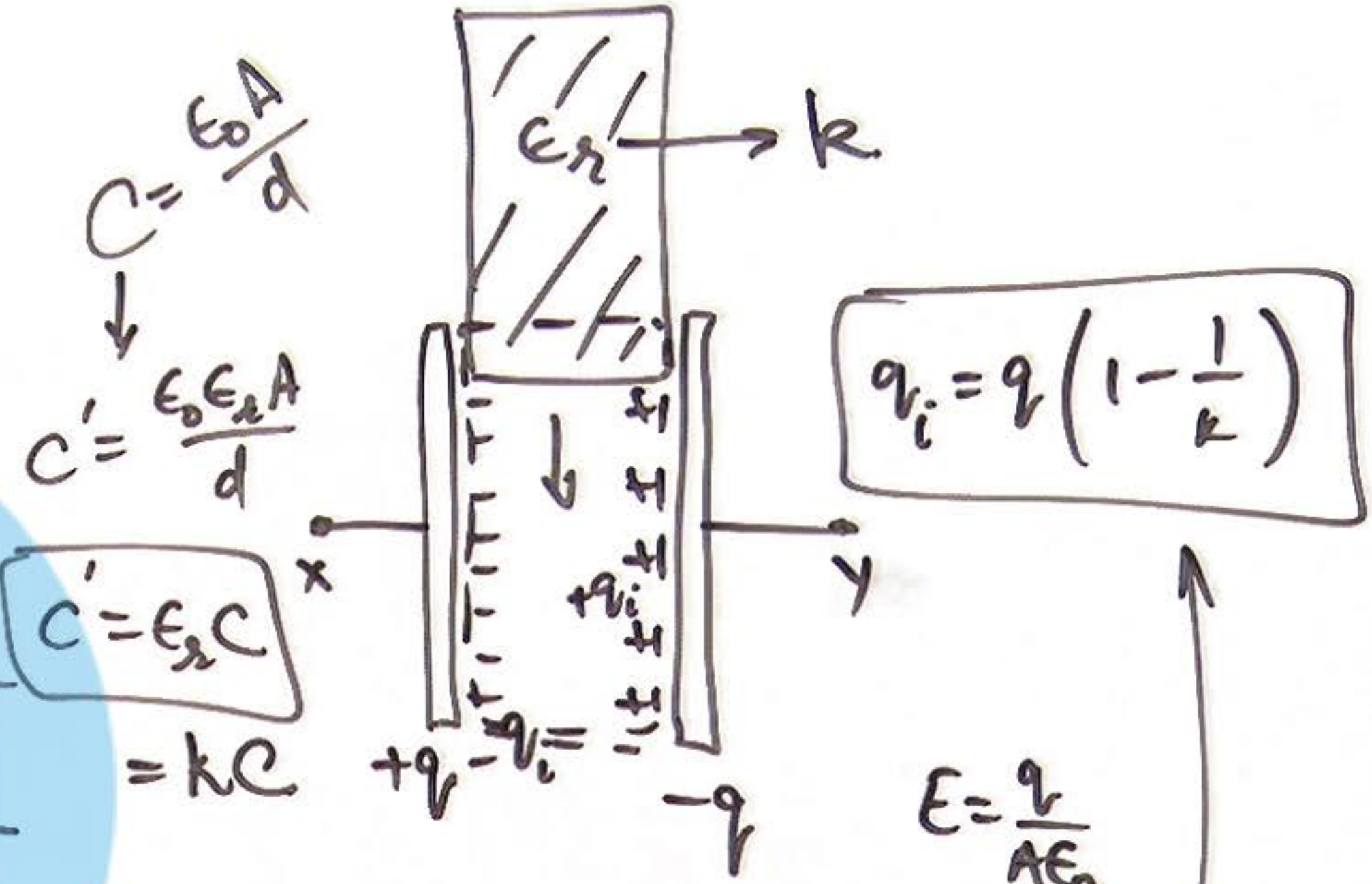
ϵ_r

$\epsilon_r = 1$ for medium

in which no polarization take place.



Physics Galaxy
Videos
Dielectrics
Slab



$$q_i = q \left(1 - \frac{1}{k} \right)$$

$$E = \frac{q}{A \epsilon_0}$$

$$q_i = q_i / A$$

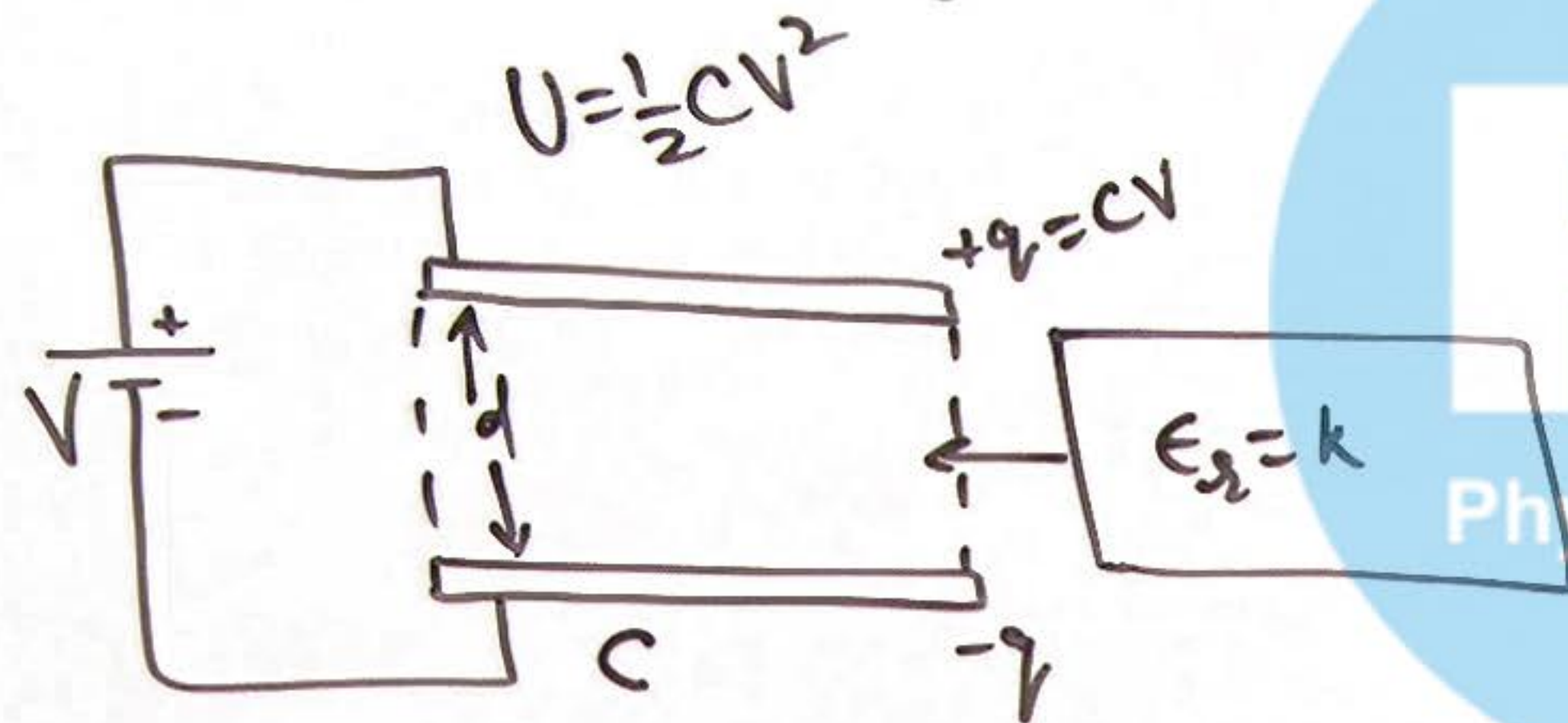
Surface density of Bound Charge

$$E_{net} = \frac{E}{\epsilon_r} = E - \frac{\sigma_i}{\epsilon_0}$$

$$\sigma_i = \epsilon_0 E \left[1 - \frac{1}{\epsilon_r} \right]$$

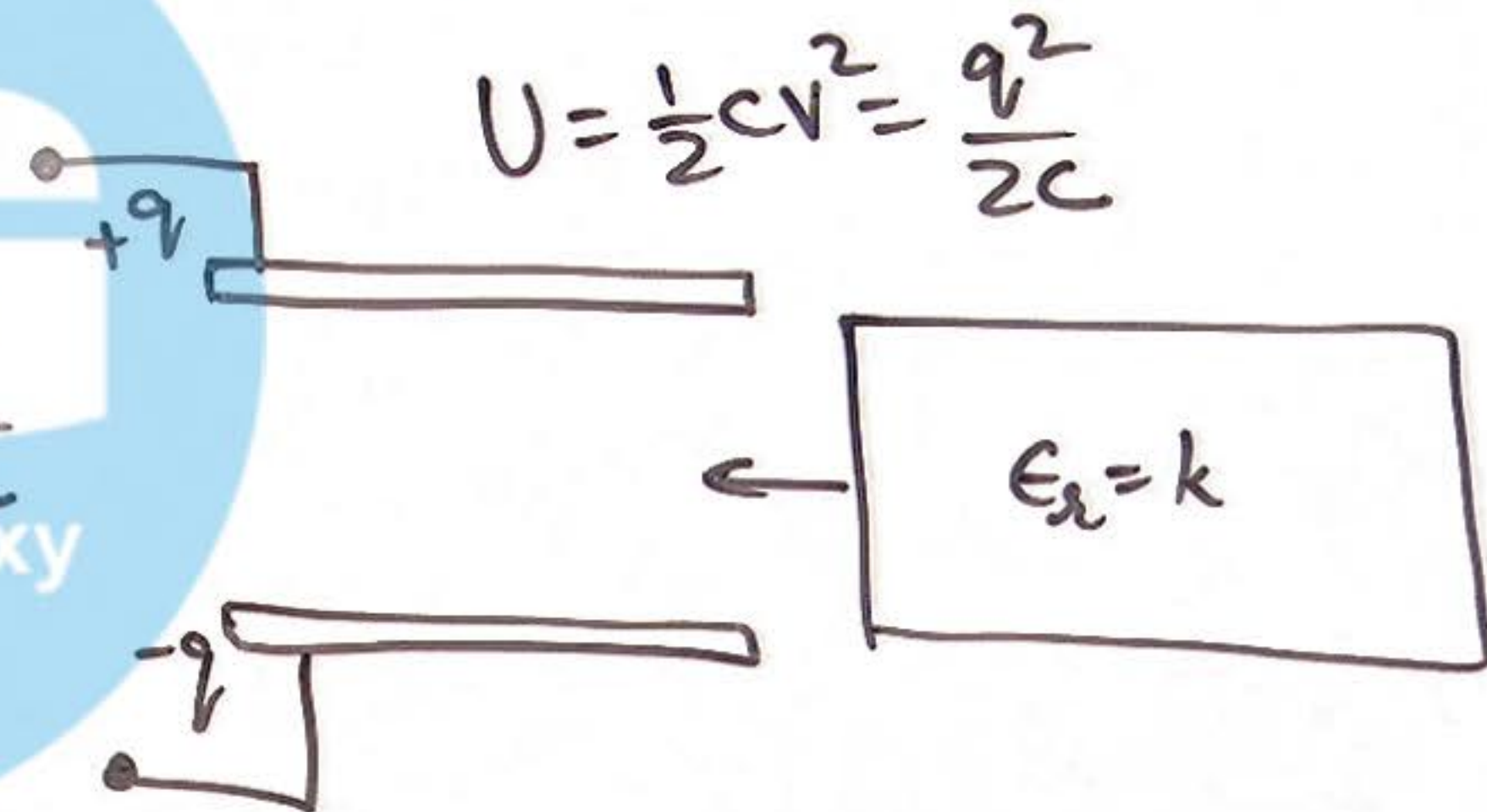
①7 Effect of Dielectric Insertion in a Capacitor:

Case-I: Battery remain Connected during insertion



Before	$\rightarrow$	q	C	V	E	U
		$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$
After	$\rightarrow$	kq	kC	V	E	kU

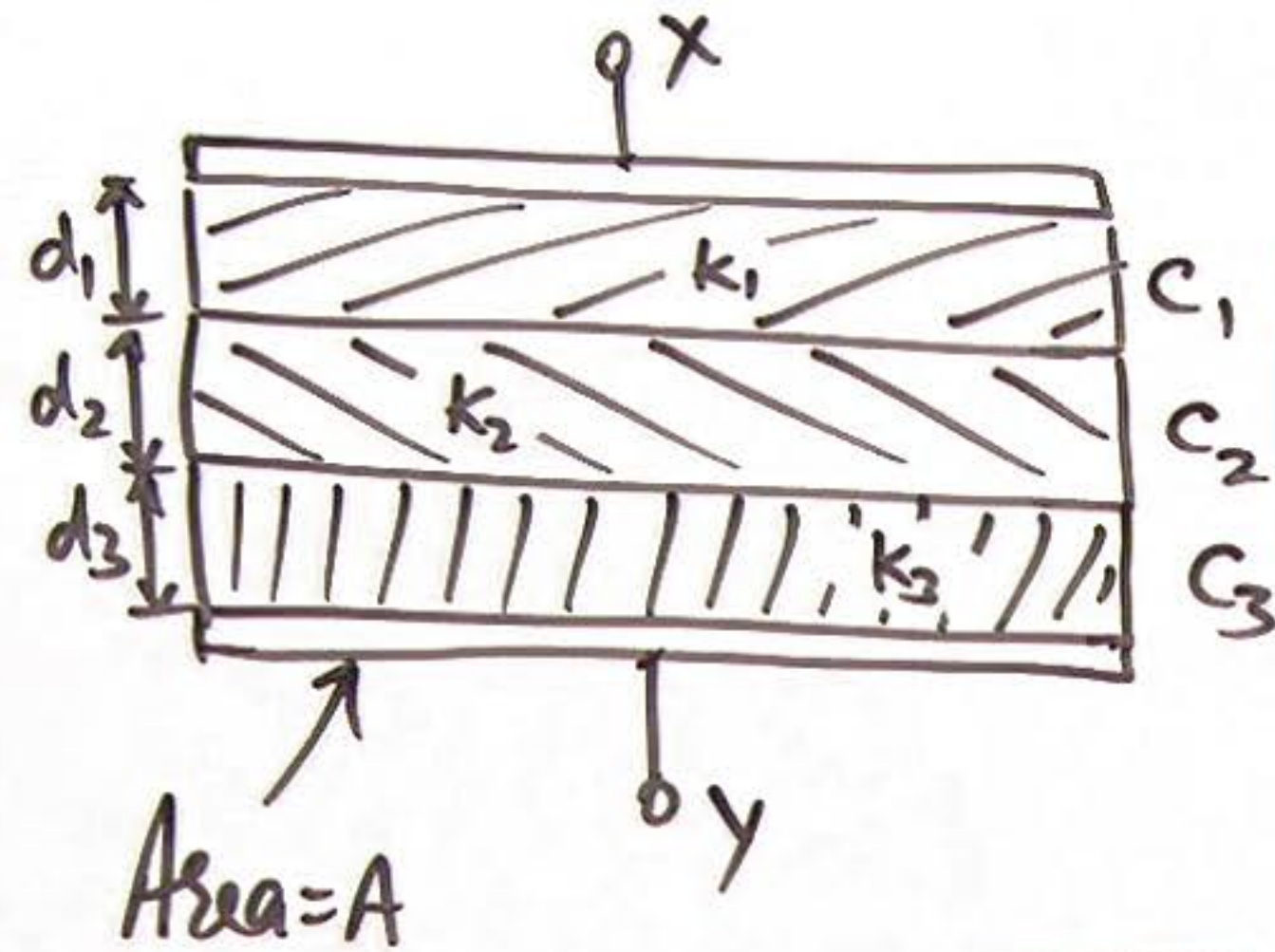
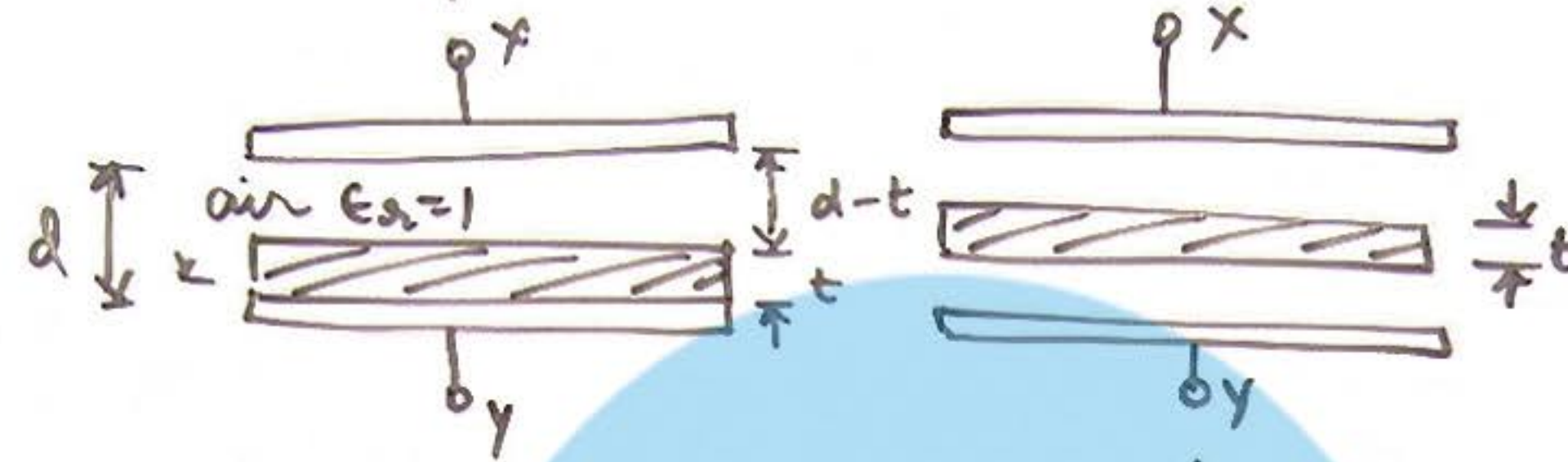
Case-II: Battery is disconnected before insertion



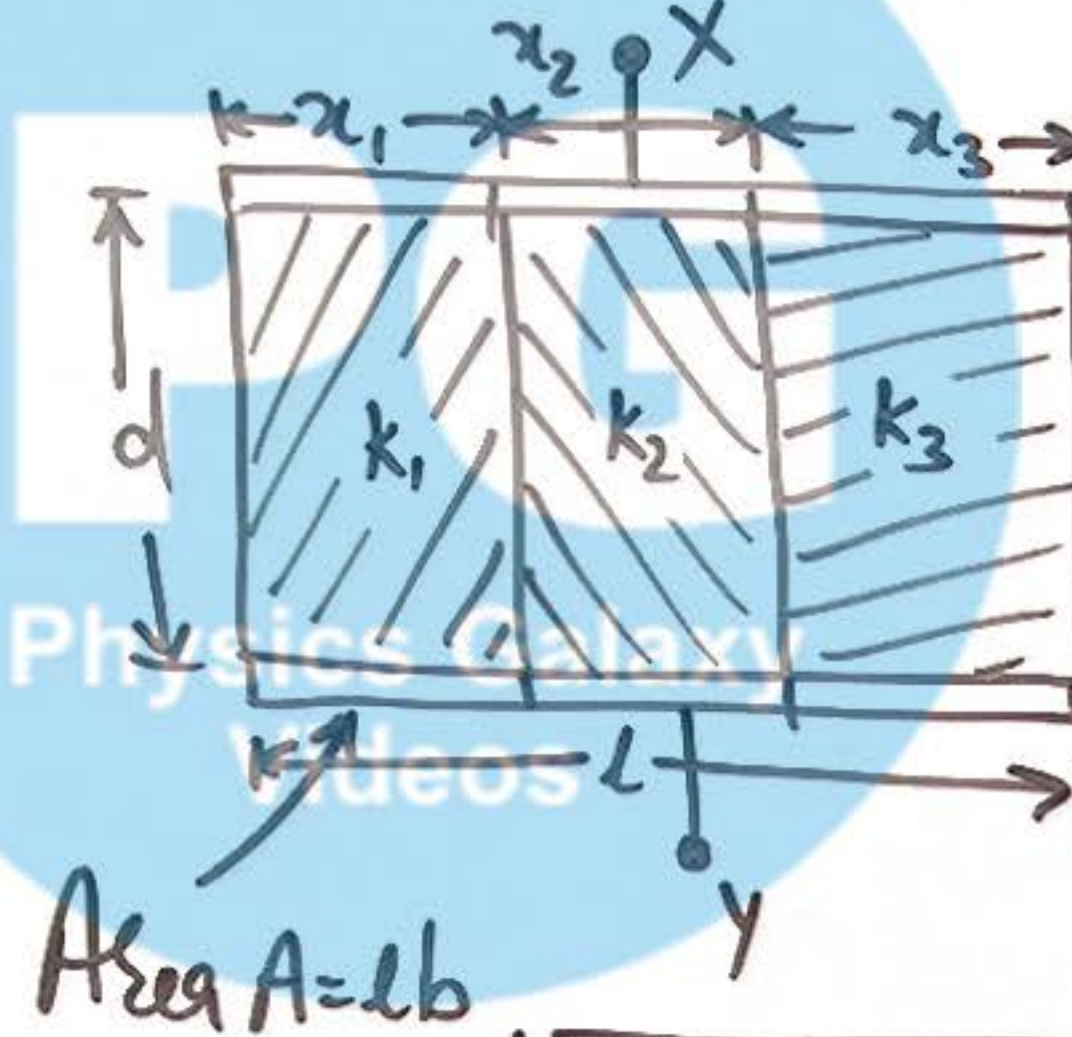
Before	$\rightarrow$	q	C	V	E	U
		$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$
After	$\rightarrow$	q	kC	V/k	E/k	U/k

18 Multiple Dielectrics in One Capacitor:

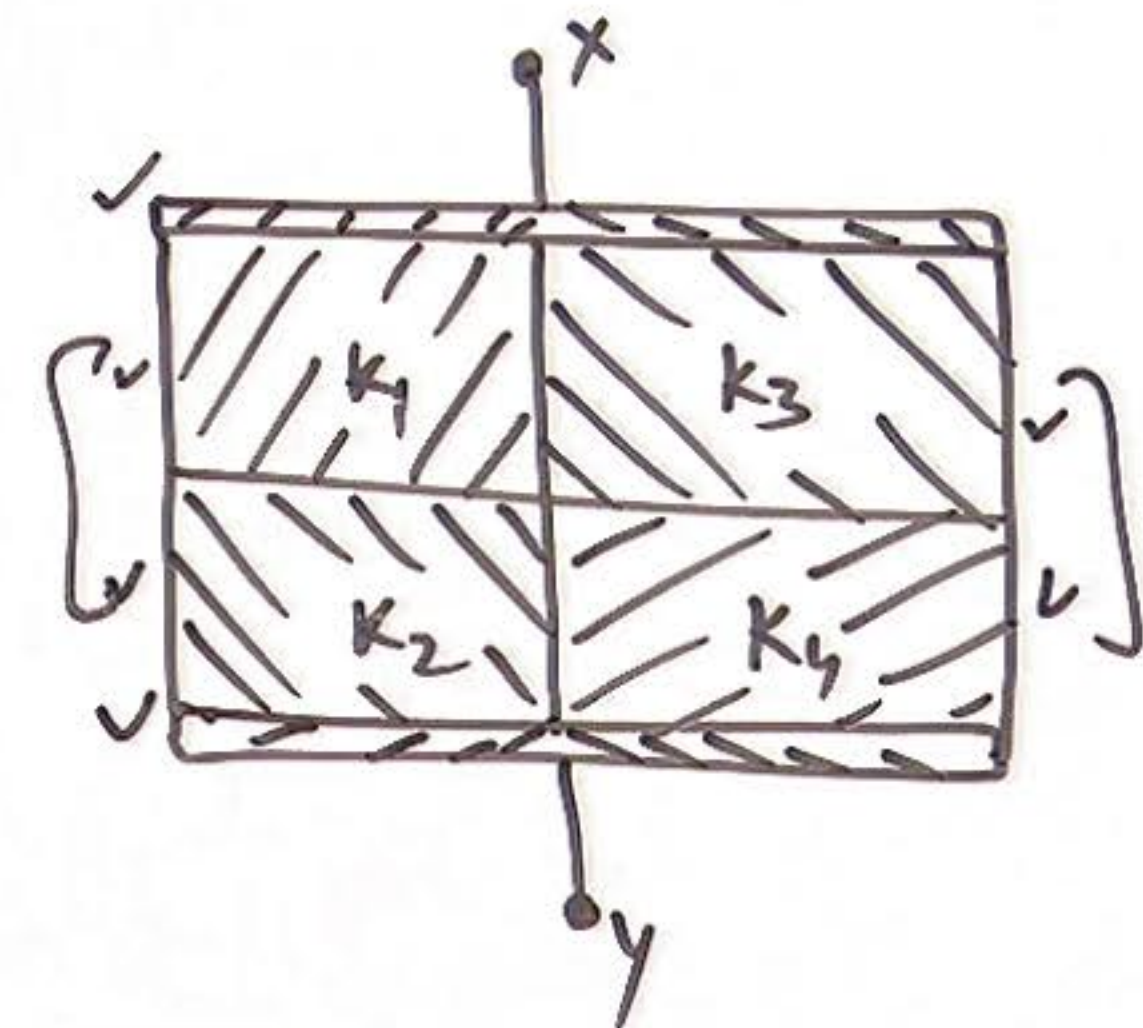
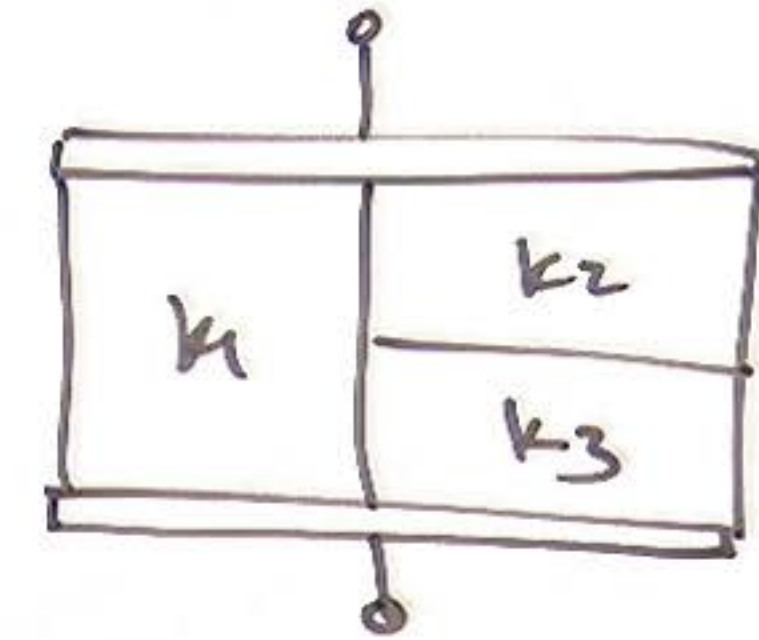
$$C_{xy} = \frac{\epsilon_0 A}{\frac{t}{k} + d - t} \checkmark$$



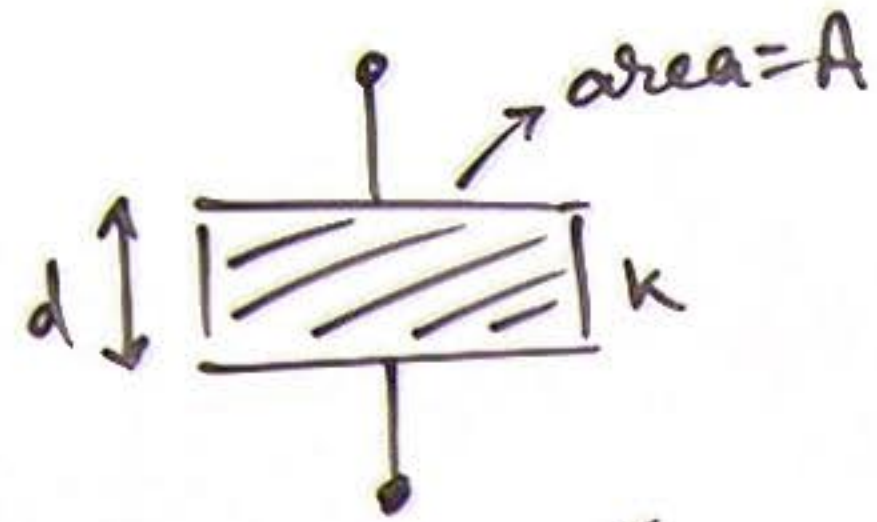
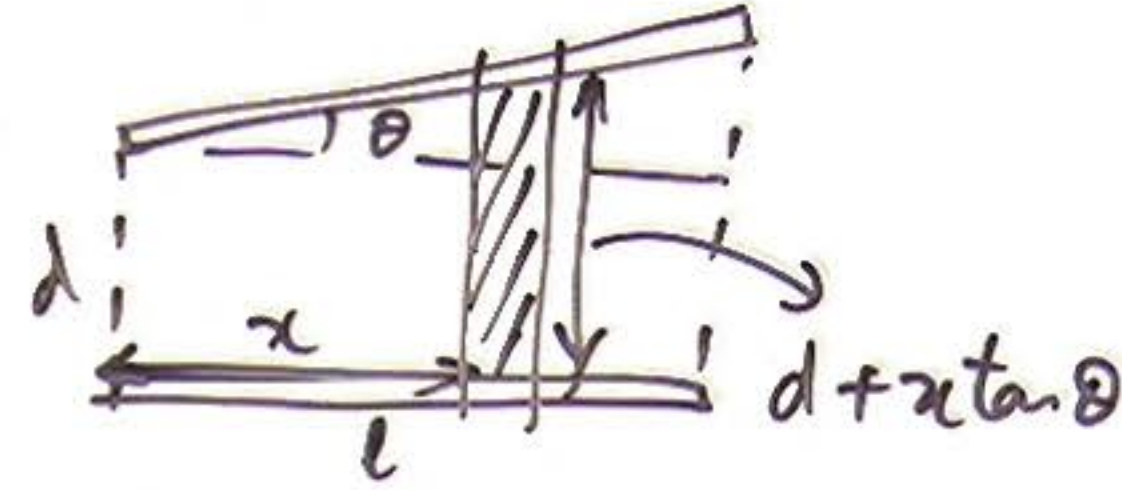
$$C_y = \frac{\epsilon_0 A}{\frac{d_1}{k_1} + \frac{d_2}{k_2} + \frac{d_3}{k_3}}$$



$$C_{xy} = \frac{k_1 \epsilon_0 x_1 b}{d} + \frac{k_2 \epsilon_0 x_2 b}{d} + \frac{k_3 \epsilon_0 x_3 b}{d}$$

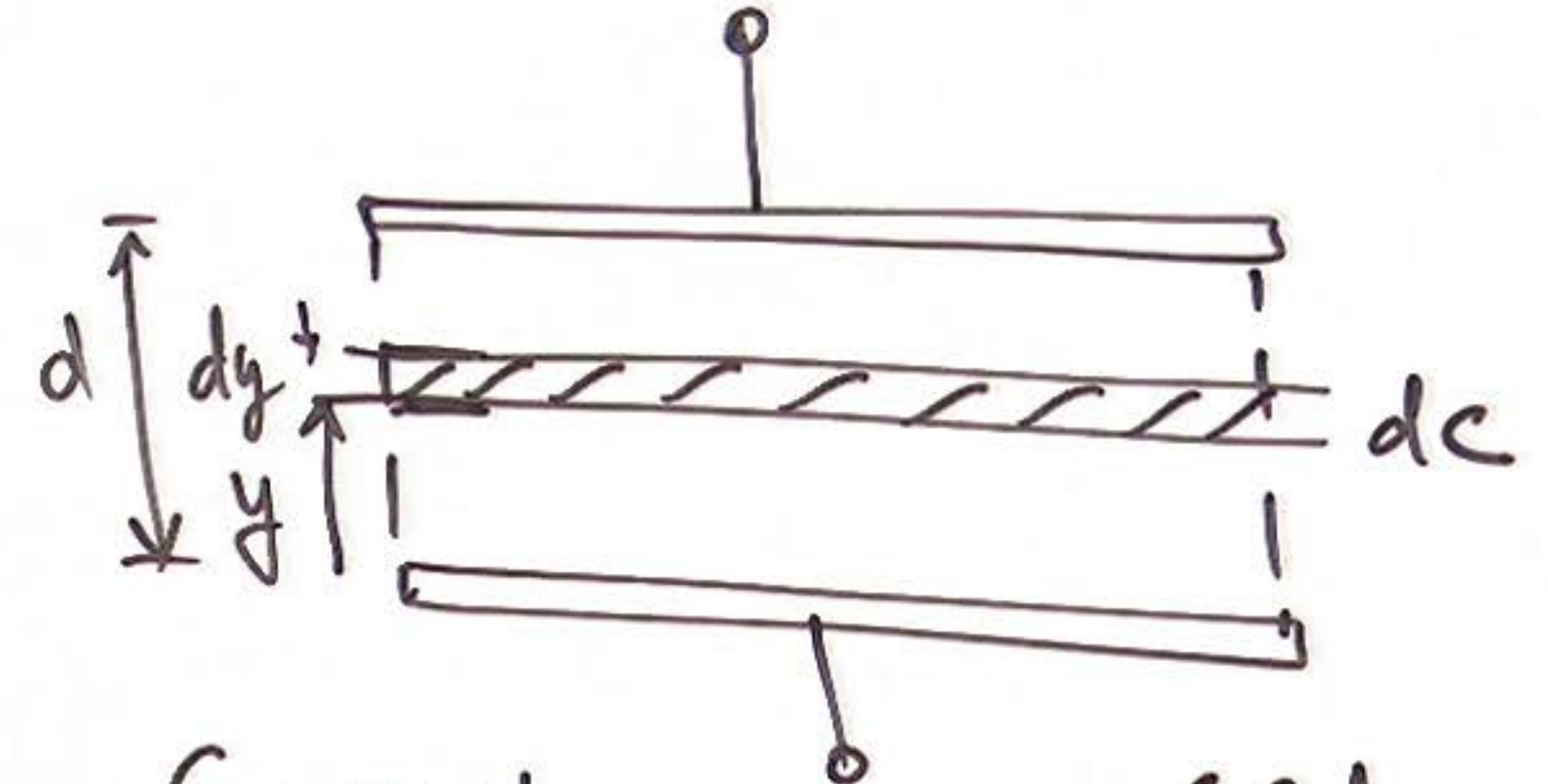
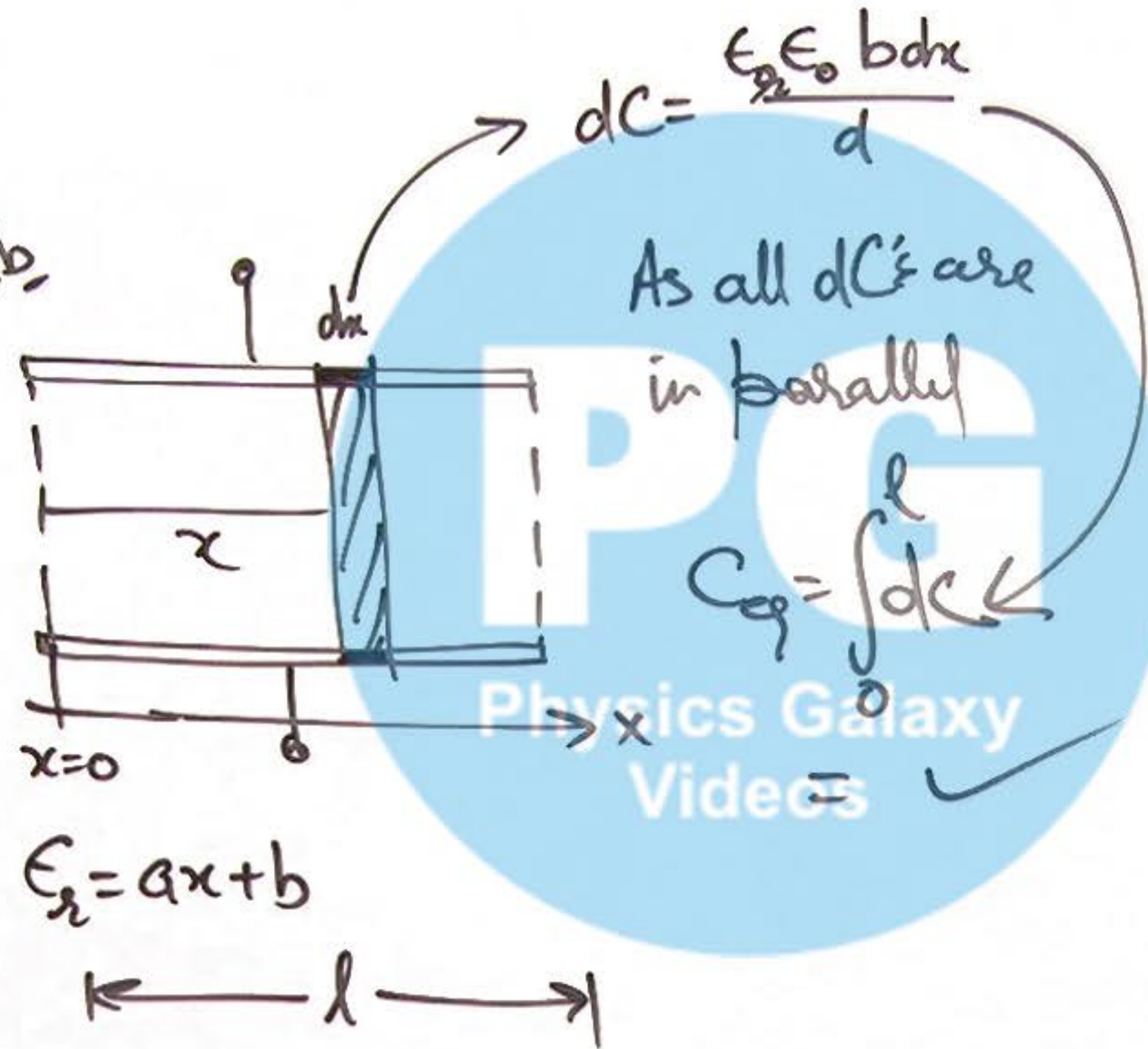
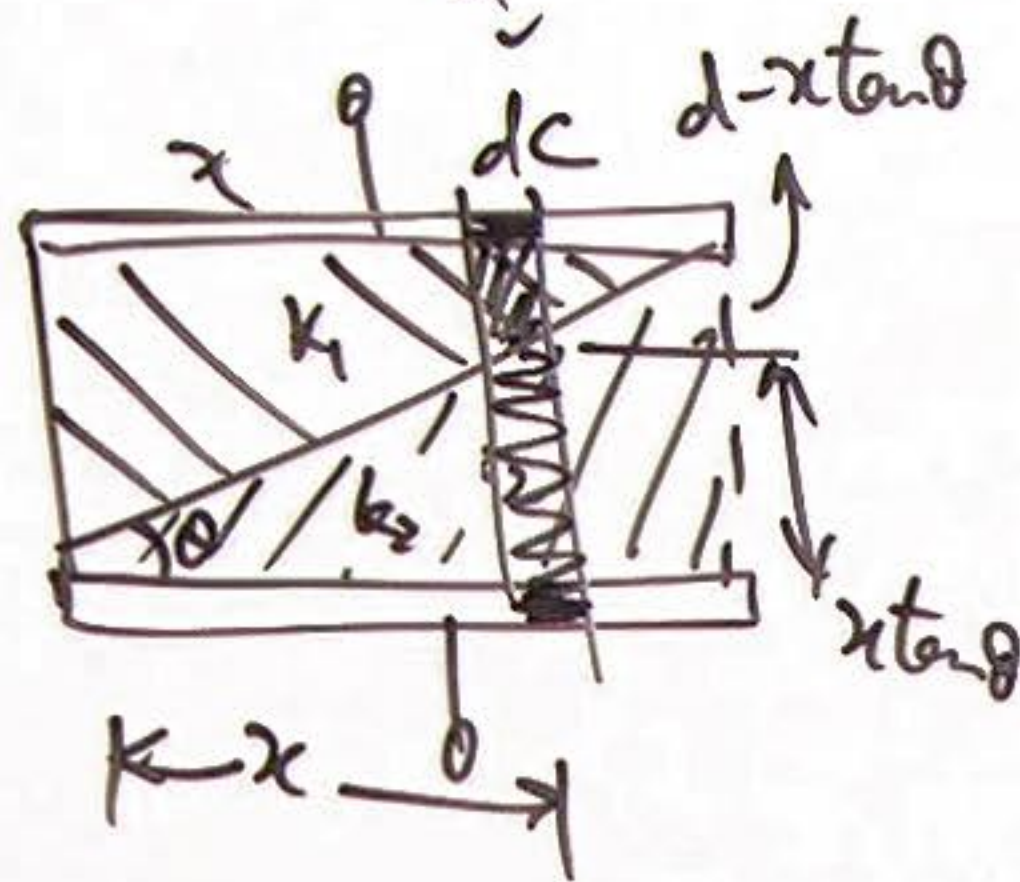


19 Capacitance with varying parameters:



$$C = \frac{k\epsilon_0 A}{d}$$

$$A = lb$$



$$\epsilon_r = ay + b$$

$$dC = \frac{\epsilon_r \epsilon_0 A}{dy}$$

As all dC s are in series

$$\Rightarrow \frac{1}{C_{eq}} = \int \frac{1}{dC}$$